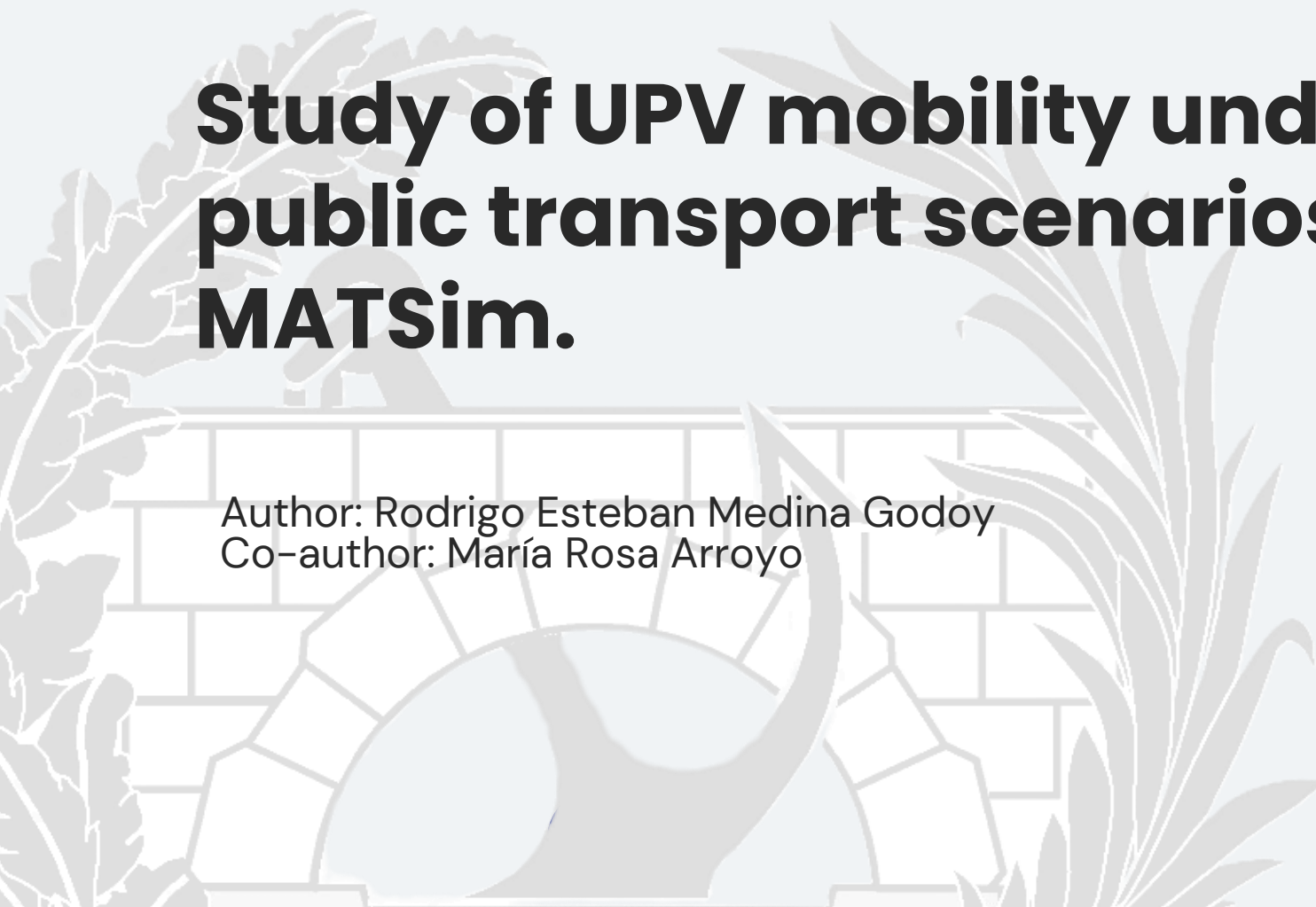


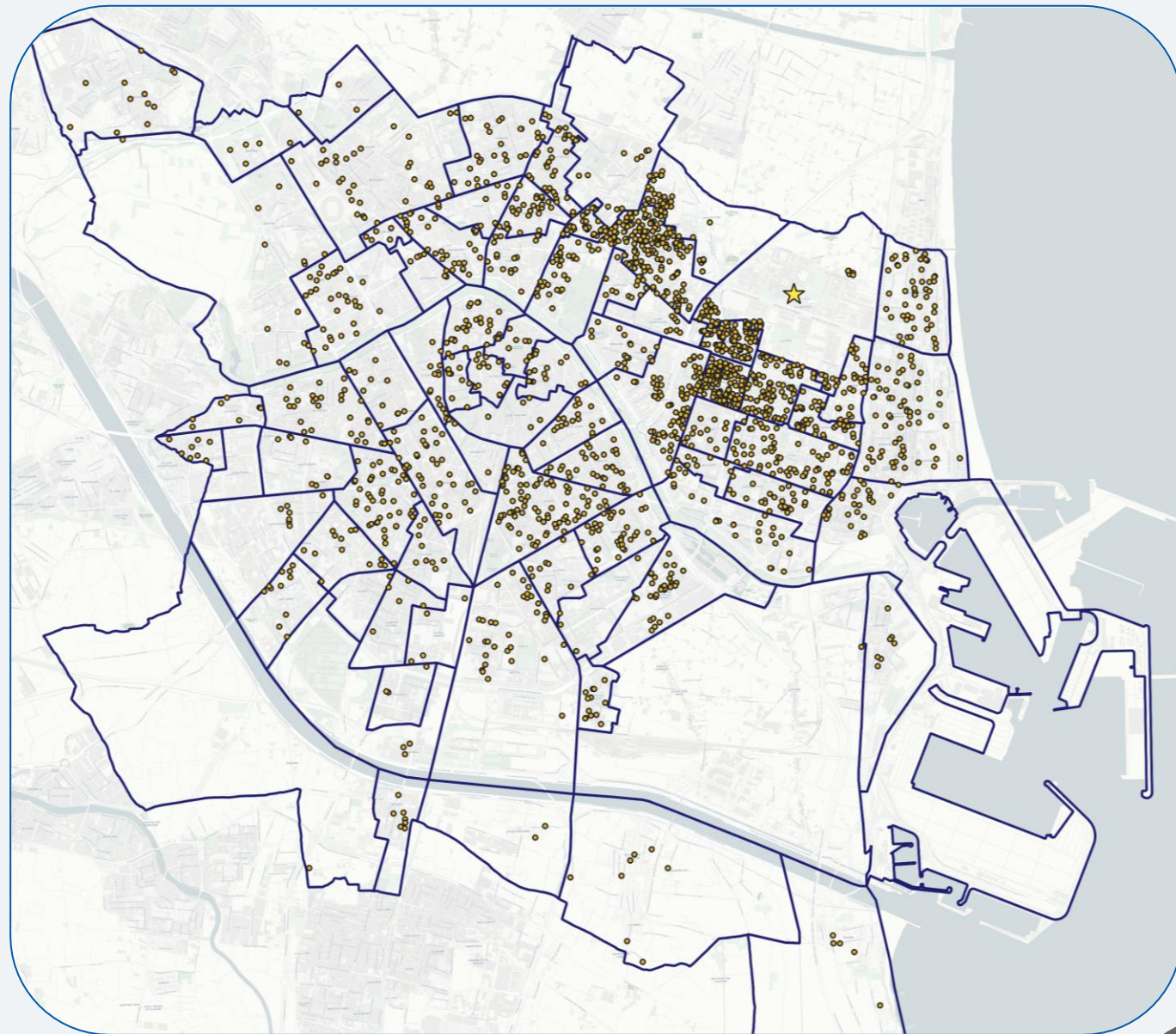
Study of UPV mobility under restrictive public transport scenarios using MATSim.

Author: Rodrigo Esteban Medina Godoy
Co-author: María Rosa Arroyo



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1. Introduction
2. Scope and Limitations
3. Methodology
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5. Model Development, Adjustment,
and Calibration
6. Results
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8. Conclusions



1.- Introduction



- Development of a new project focused on studying transport disruptions caused by moderate flooding events.
- A fully functional and calibrated model for the Valencia Metropolitan Area is currently under development.
- This project includes certain limitations and adjustments compared to the initial development plan and the requested specifications. Therefore, it does not include a synthetic population for the entire city or mobility modelling at the metropolitan level.



Valencia, Spain
September 16 - 17th,
2026

2.- Scope and Limitations



- **Mobility** exclusively within **Valencia city**.
- The simulated population is based on the May **2022 UPV Mobility Survey**.
- **Only** urban **public transport data** (EMT and FGV) are considered.
- All **data** used correspond to the **2022** time **horizon**.



- There is a **limitation** in the creation of detailed plans due to the **survey's structure**.
- Simulating **only** UPV **users** does not generate real **competition** for road **capacity**, although it accurately reflects flows to and from the campus.
- **Activities** were **simplified** according to the user group.
- A **statistical simplification** was applied for responses involving **transfers** or **multiple** transport **modes**.



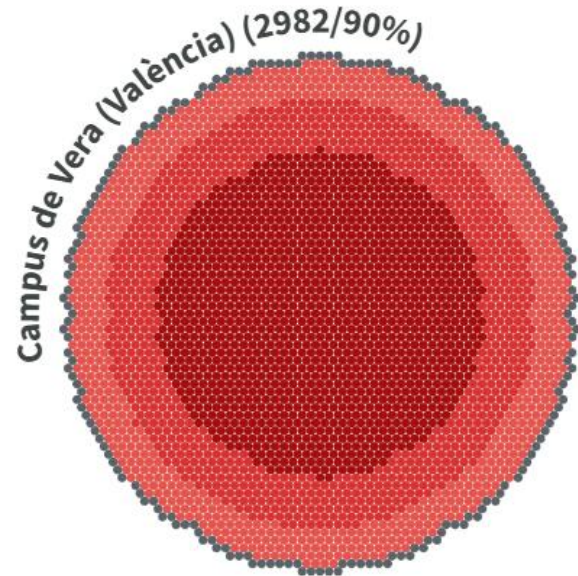
3.- Methodology

- Comprehensive cleaning, filtering, and processing of data extracted from the UPV mobility survey.
 - **Spatial processing** using QGIS to establish the agents' spatial references.
 - **Python programming** to transform the processed survey data into the specific format required by MATSim (.xml).
 - **Manual creation of restrictive scenarios** to progressively remove public transport lines.
 - Use of a Python script to modify the plans.xml file, randomly reassigning affected public transport users to bicycles.
- 

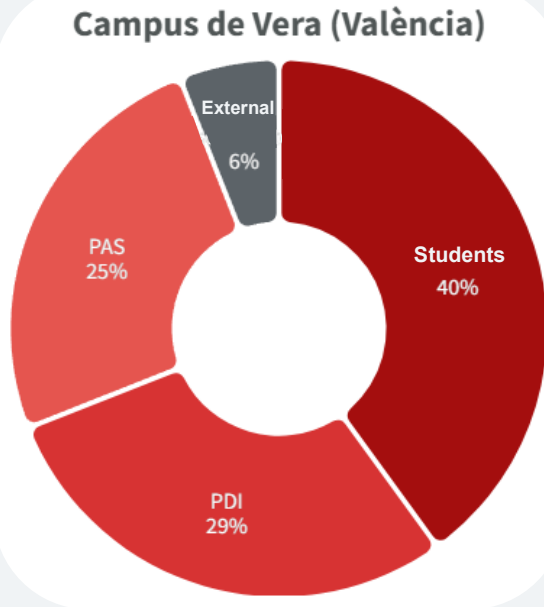
4.- Survey Mobility UPV (may 2022)

Survey Results

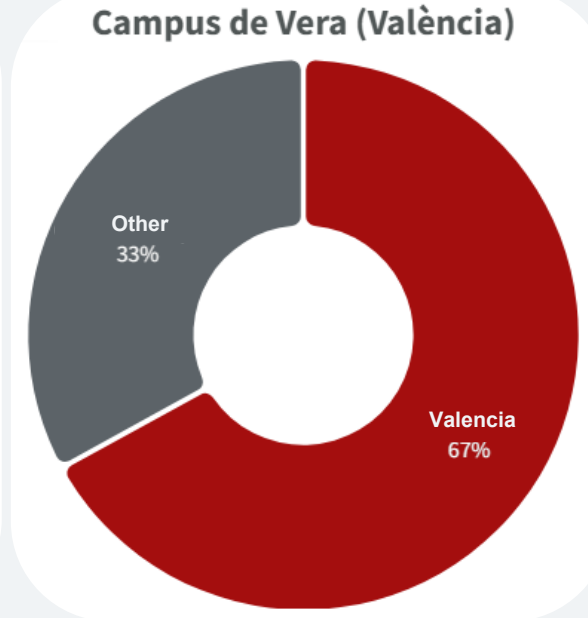
- Students
- PAS
- PDI
- External Workers



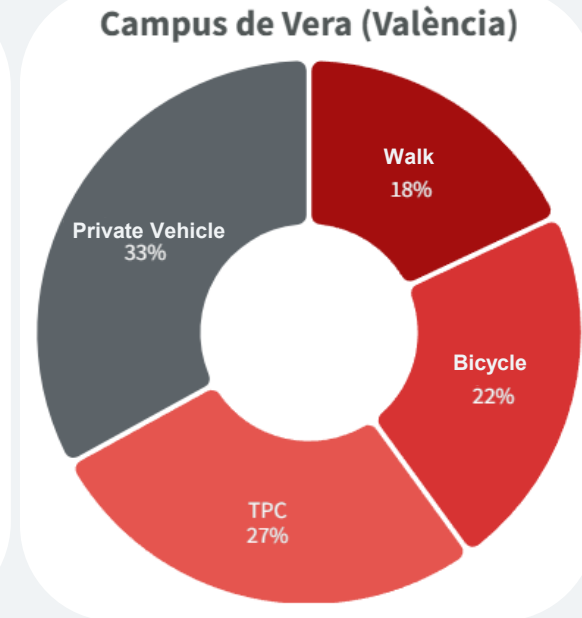
Colective



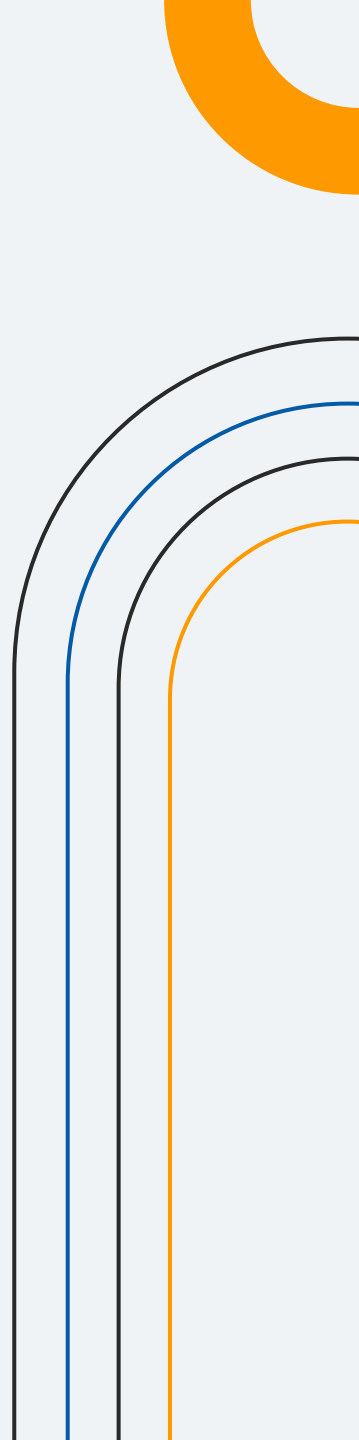
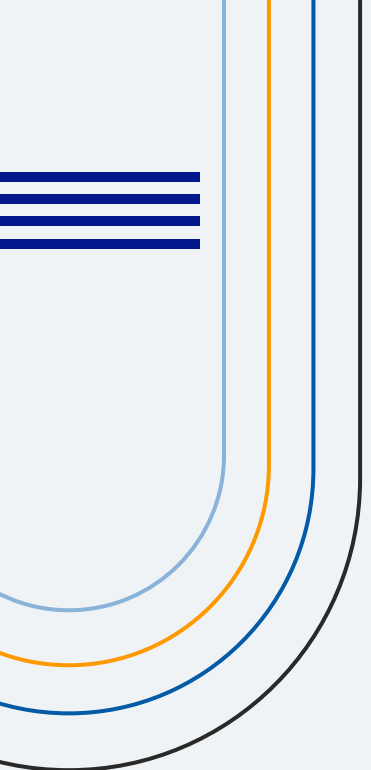
Origin



Modes



University community = 33.000 (students + personal)



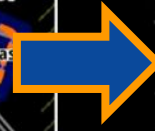
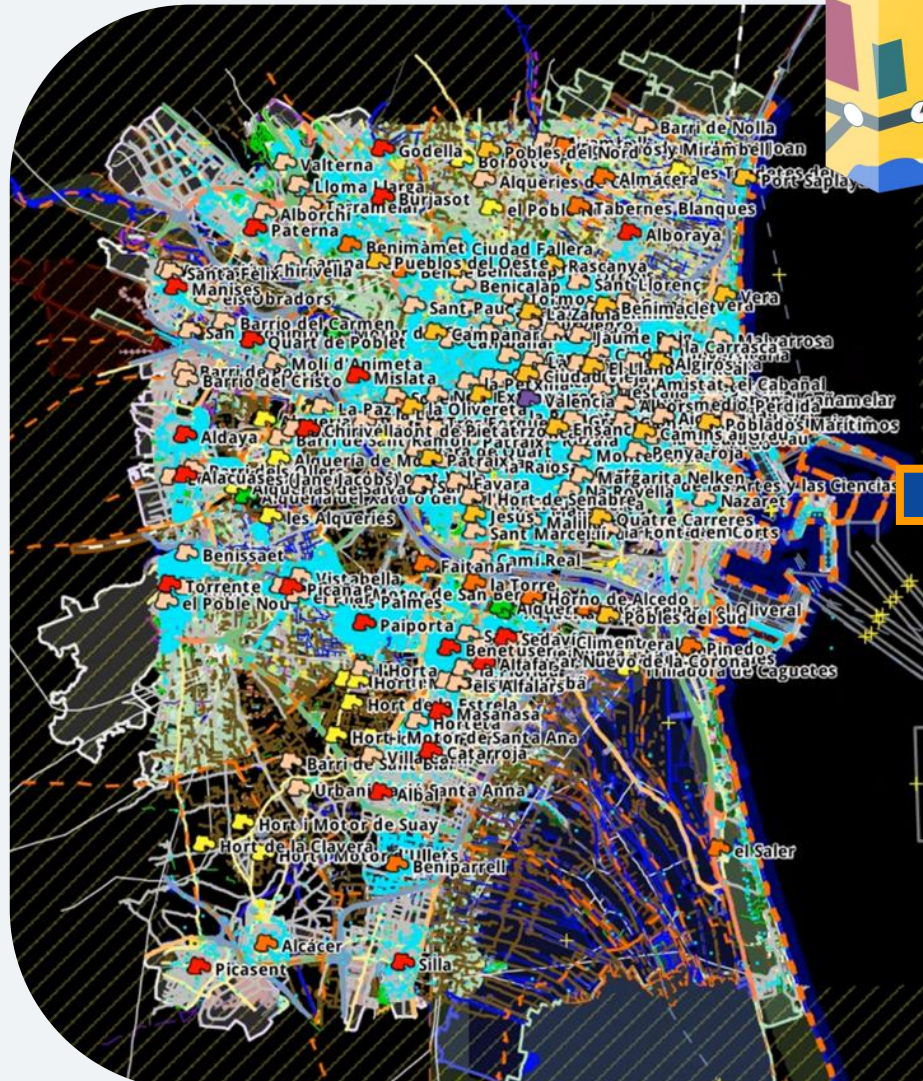
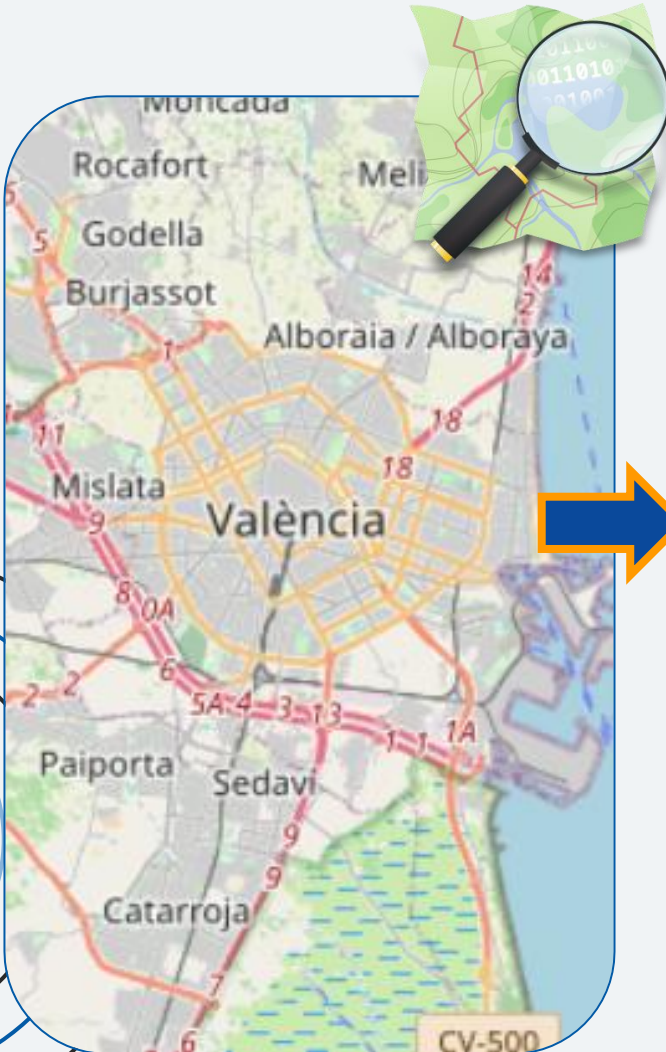
5

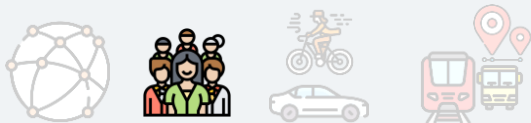
Model Development, Adjustment, and Calibration



Network.xml

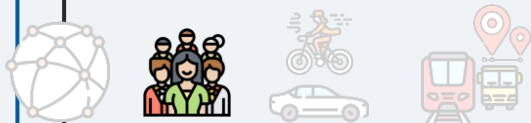
- Extraction of .osm base data from OpenStreetMap via bbbike.org
- Adjustment of UTM coordinates for Valencia (EPSG:25830)
- Conversion to MATSim .xml format using JOSM



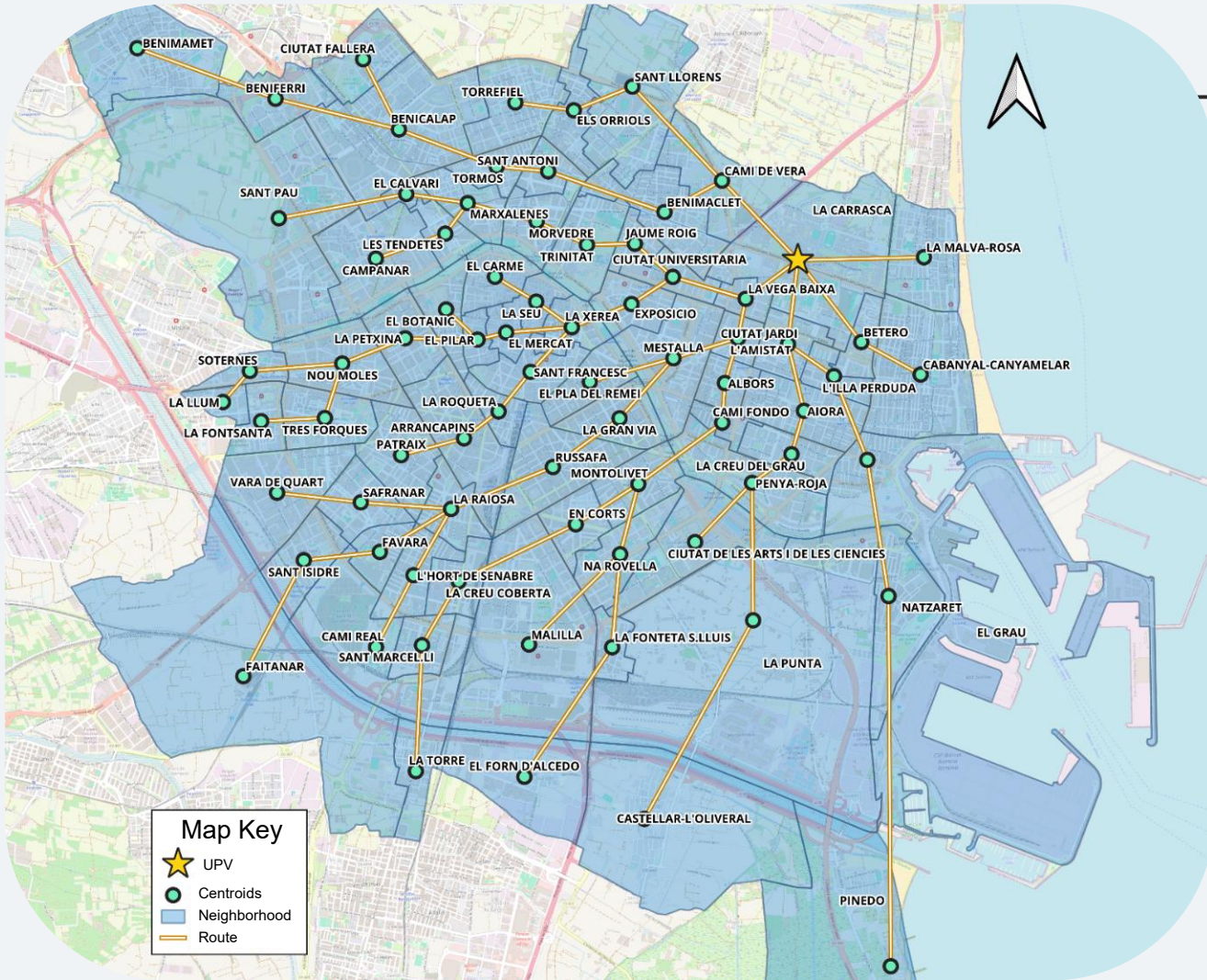
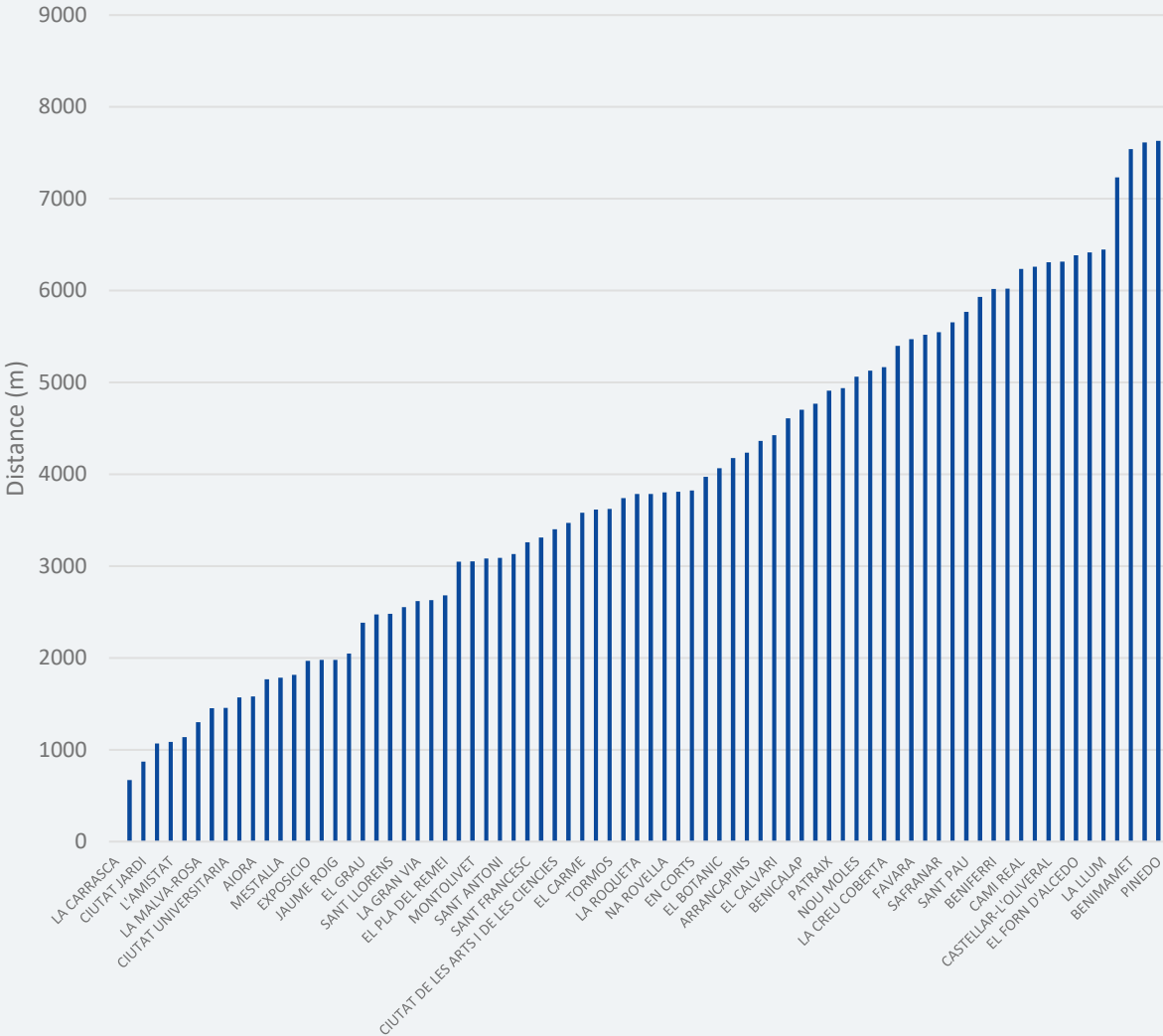


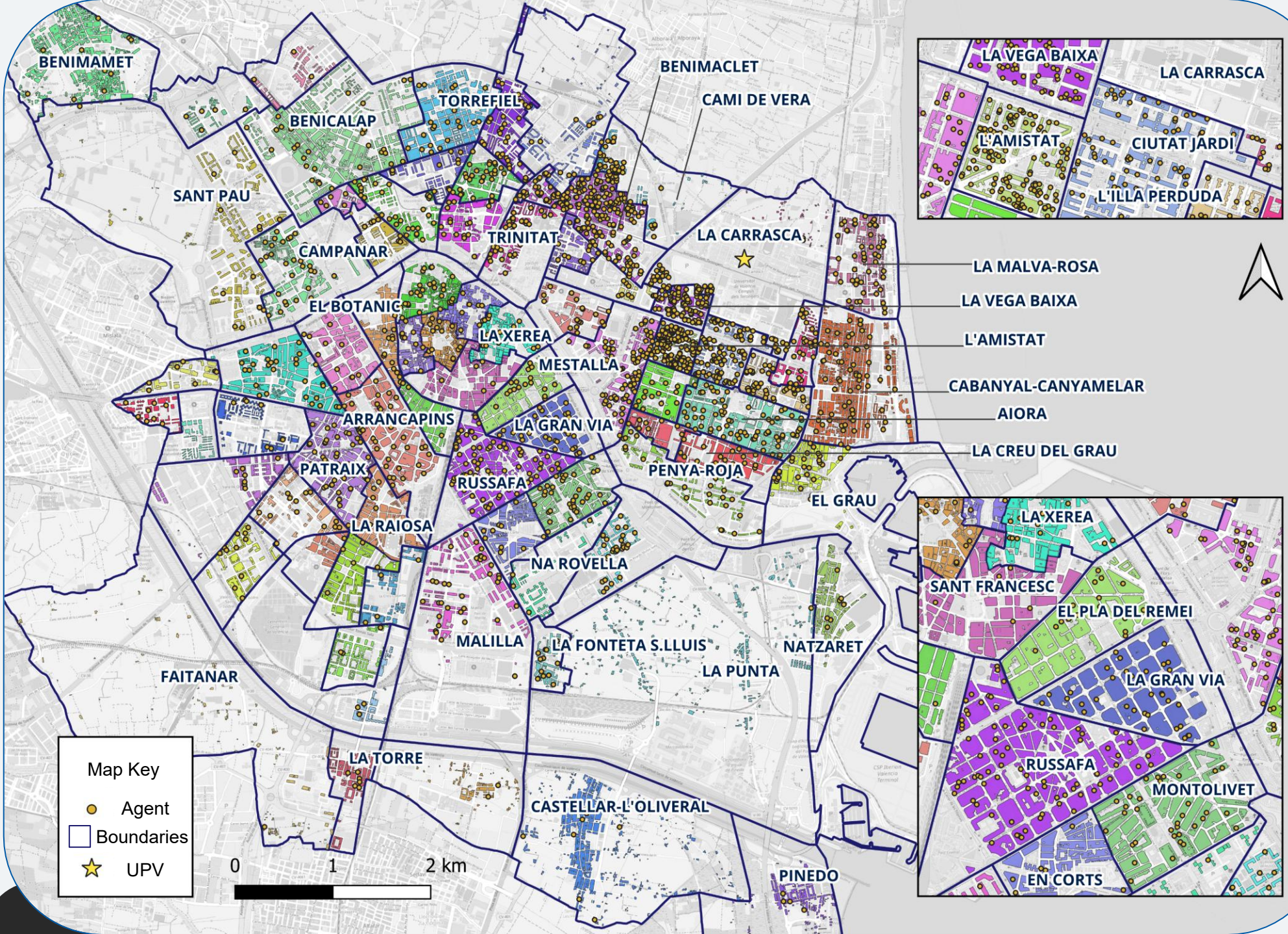
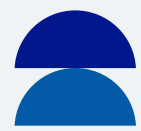
Plans.xml

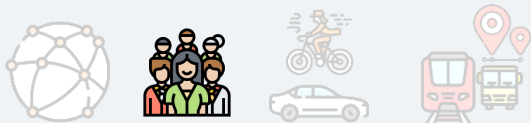
- Survey data filtered for use as the basis for plan generation.
- Limited to four modes: car, public transport (PT), bicycle, and walking.
- Responses indicating multiple modes were reduced to a single mode by applying probabilities based on distance and the Monte Carlo statistical method.
- Respondents who selected "car" plus another mode were assigned the attribute "carAvail=always," allowing them to use the car during iterations involving mode reassignment; the attribute "never" was assigned to the rest.
- The exact departure time from home was estimated by subtracting the theoretical travel time from the arrival time indicated in the survey.
- Randomly selected origin points located strictly within the residential polygons of each neighborhood.
- All information consolidated into a CSV file was processed using Python code to generate the hierarchical structure of the *plans.xml* file required by MATSim.



Distance between neighborhood centroids and “La Carrasca”







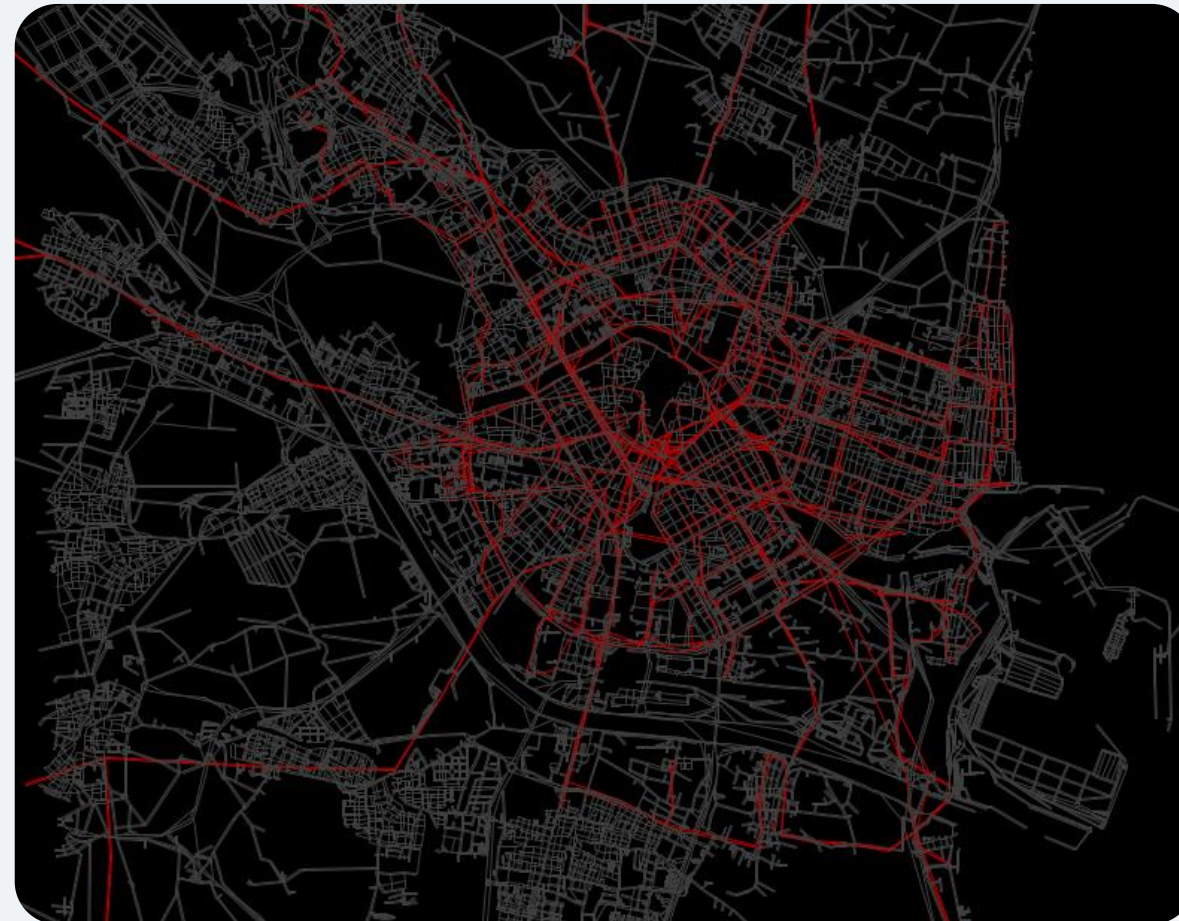
```
<person id=17>
  <attributes>
    <attribute name=carAvail class=java.lang.String>never</attribute>
  </attributes>
  <plan selected=yes>
    <activity type=h x=724924.7821 y=4371420.408 end_time=7:03:30/>
    <leg mode=bicycle/>
    <activity type=s x=728351.9221 y=4373419.186 end_time=20:00:00/>
    <leg mode=bicycle/>
    <activity type=h x=724924.7821 y=4371420.408/>
  </plan>
</person>
```

id	home_x	home_y	upv_x	upv_y	activity	mode	Departure time_1	Departure time_2	Car_Avail
17	724924.782	4371420.41	728351.922	4373419.19	s	bicycle	7:03:30	20:00:00	never
20	724265.614	4371129.9	728439.803	4373393.52	w	car	8:54:33	18:00:00	always
22	725605.712	4375248.42	728773.947	4373277.13	w	car	7:42:45	19:00:00	always
23	726037.031	4373588.69	728276.753	4373445.16	s	walk	8:52:58	18:00:00	never
25	723459.688	4370664.82	728178.521	4373477.91	w	car	7:42:12	18:00:00	always
26	727188.073	4374127.6	728614.591	4373328.97	w	bicycle	9:35:48	18:00:00	always
28	727484.794	4372016.09	728178.521	4373477.91	w	pt	7:52:46	17:00:00	always
29	729716.347	4373031.58	728439.803	4373393.52	w	bicycle	8:56:18	15:00:00	always
30	725654.049	4374822.53	728516.017	4373366.3	w	pt	9:10:07	17:00:00	never
32	728270.178	4371381.55	728351.922	4373419.19	w	bicycle	7:50:33	19:00:00	never
34	726196.752	4373500.78	728516.017	4373366.3	s	bicycle	6:52:58	18:00:00	never



TransitSchedule.xml & TransitVehicles.xml

- GTFS data transformation using the GTFS2MATSim GitHub repository; this enables the generation of the TransitSchedule.xml and the public transport sub-network.
- The TransitVehicles.xml (fleet) represents the supply, defining the number of vehicles, IDs, capacity, length, and the equivalent space occupied on the road (PCE).



PT



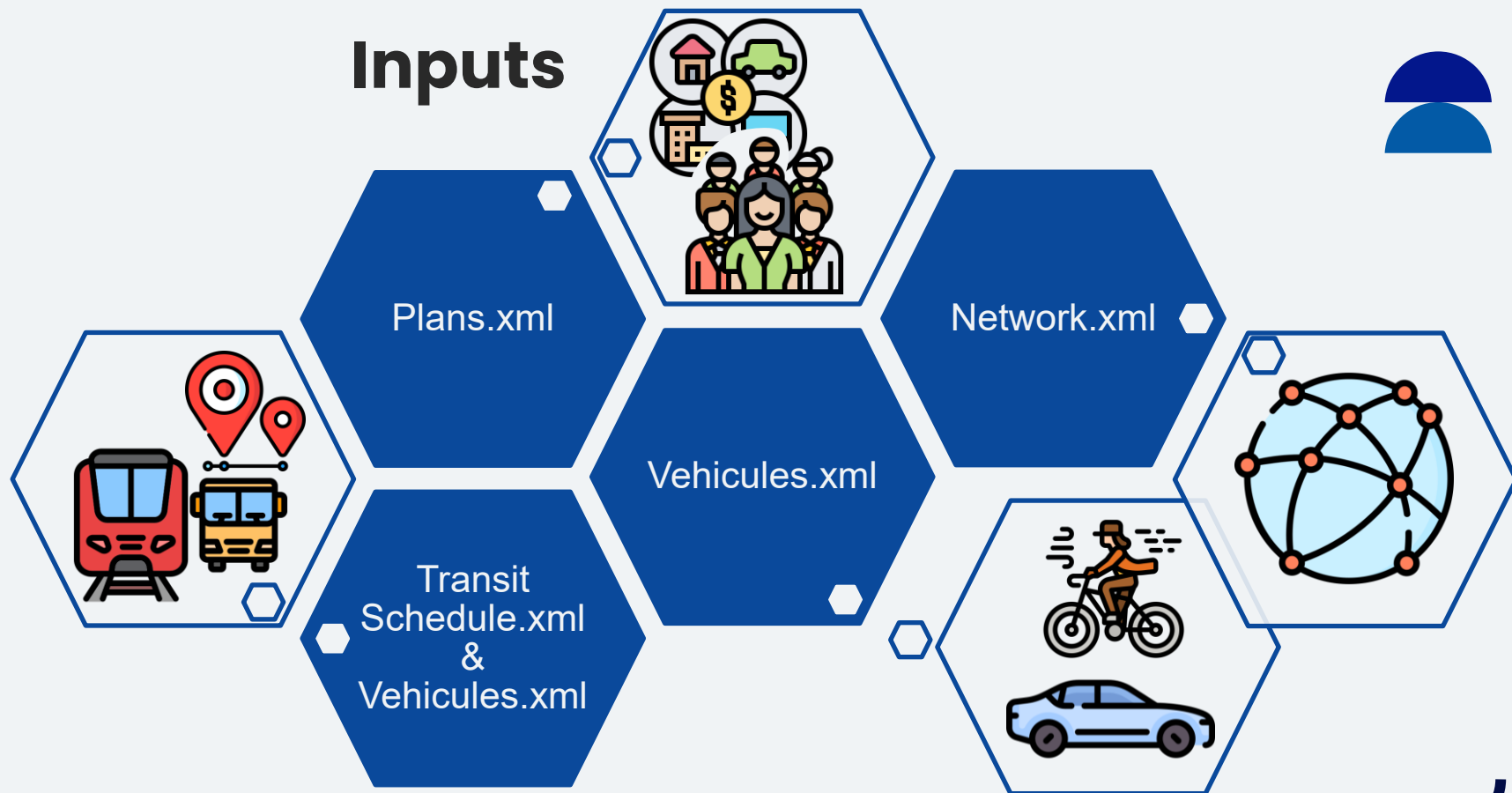
Vehicle type	Fabricante	Capacity			Dimensions		Operation Times per Pax			PCE
		Total Capacity	Seats	Stand	Length	Width	Boarding (s)	Alighting (s)	Operation type	
Metro	Vossloh	588	116	472	60.49	2.55	0.05	0.05	serial	8.0
Tram 1	Alsthom	201	65	136	23.78	2.4	0.2	0.2	serial	3.0
Tram 2	Bombardier	277	55	222	32.2	2.4	0.2	0.2	serial	4.2
Bus	Mercedes Citaro	102	28	74	12.1	2.3	0.3	0.3	parallel	1.6
Articulated bus	Mercedes Citaro	158	49	109	18	2.3	0.3	0.3	parallel	2.4

Configured Modules

Controller
Qsim
Scoring
TransitRouter
Routing
Replanning

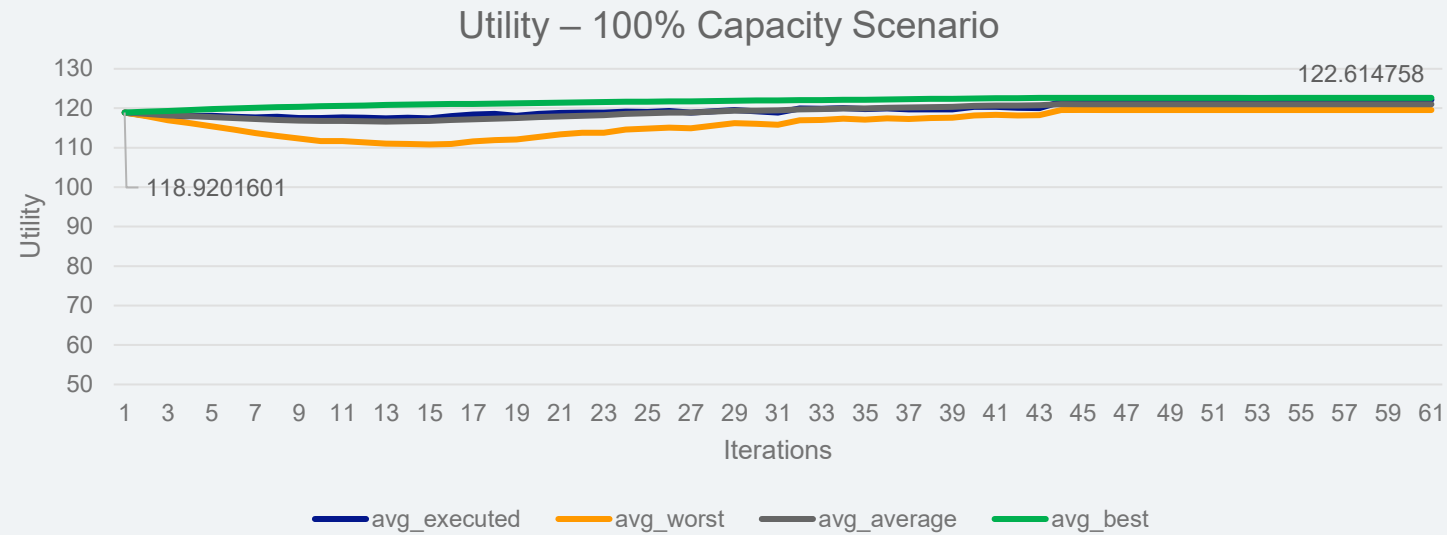


Inputs



Adjustment and Calibration

- Simulating the entire available infrastructure using only a fraction of the population eliminates congestion.
- By simulating a limited sample rather than the whole of Valencia, the calibration did not aim to match exact traffic volumes, but rather to stabilize and adjust the modal split.



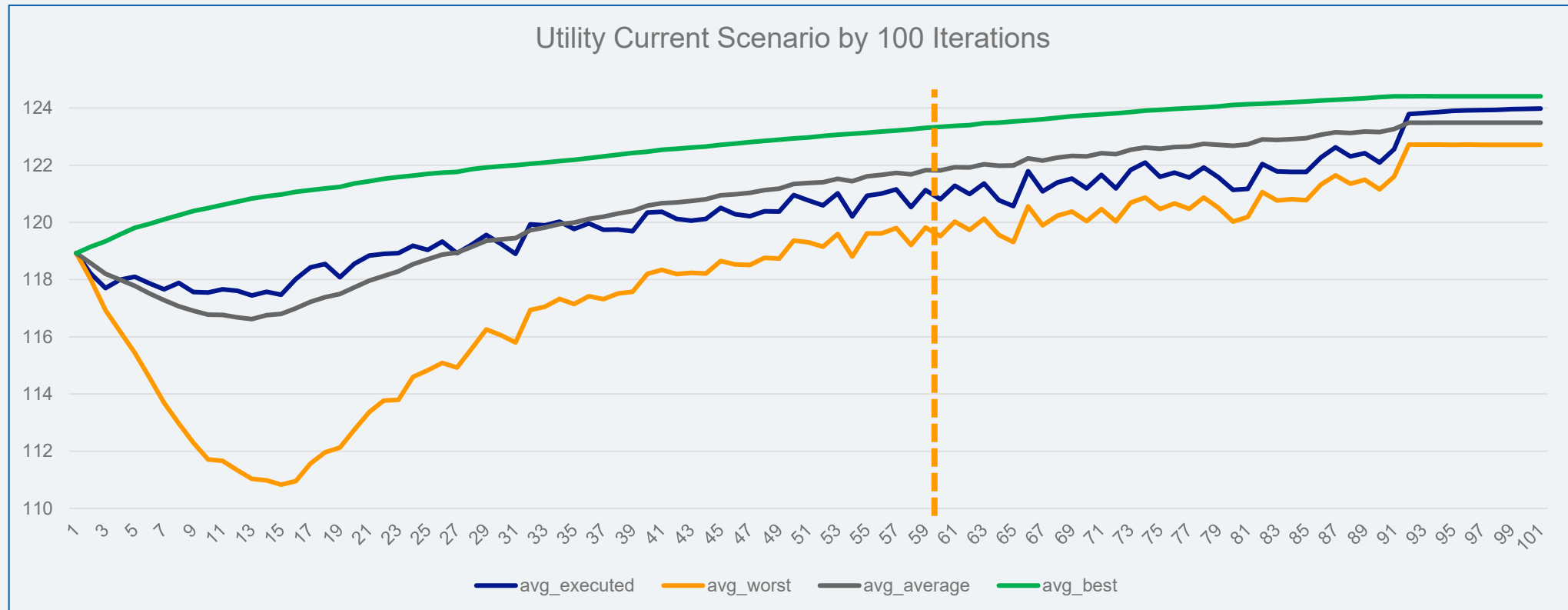
Percentages	FlowcapacityFactor	storageCapacityFactor
Scenario 5%	0.05	0.05
Scenario 4%	0.04	0.04
Scenario 3%	0.03	0.03
Scenario 2%	0.02	0.02
Scenario 1%	0.01	0.01

Vehicle Type	Total Capacity	Seats	Walk	Scenario 5%		Scenario 4%		Scenario 3%		Scenario 2%		Scenario 1%	
Metro - Vossloh	588	116	472	6	24	5	19	3	14	2	9	1	5
Tram 1 - Alsthom	201	65	136	3	7	3	5	2	4	1	3	1	1
Tram 2 - Bombardier	277	55	222	3	11	2	9	2	7	1	4	1	2
Bus - Mercedes Citaro	102	28	74	1	4	1	3	1	2	1	1	0	1
Articulated Bus - Mercedes Citaro	158	49	109	2	5	2	4	1	3	1	2	0	1

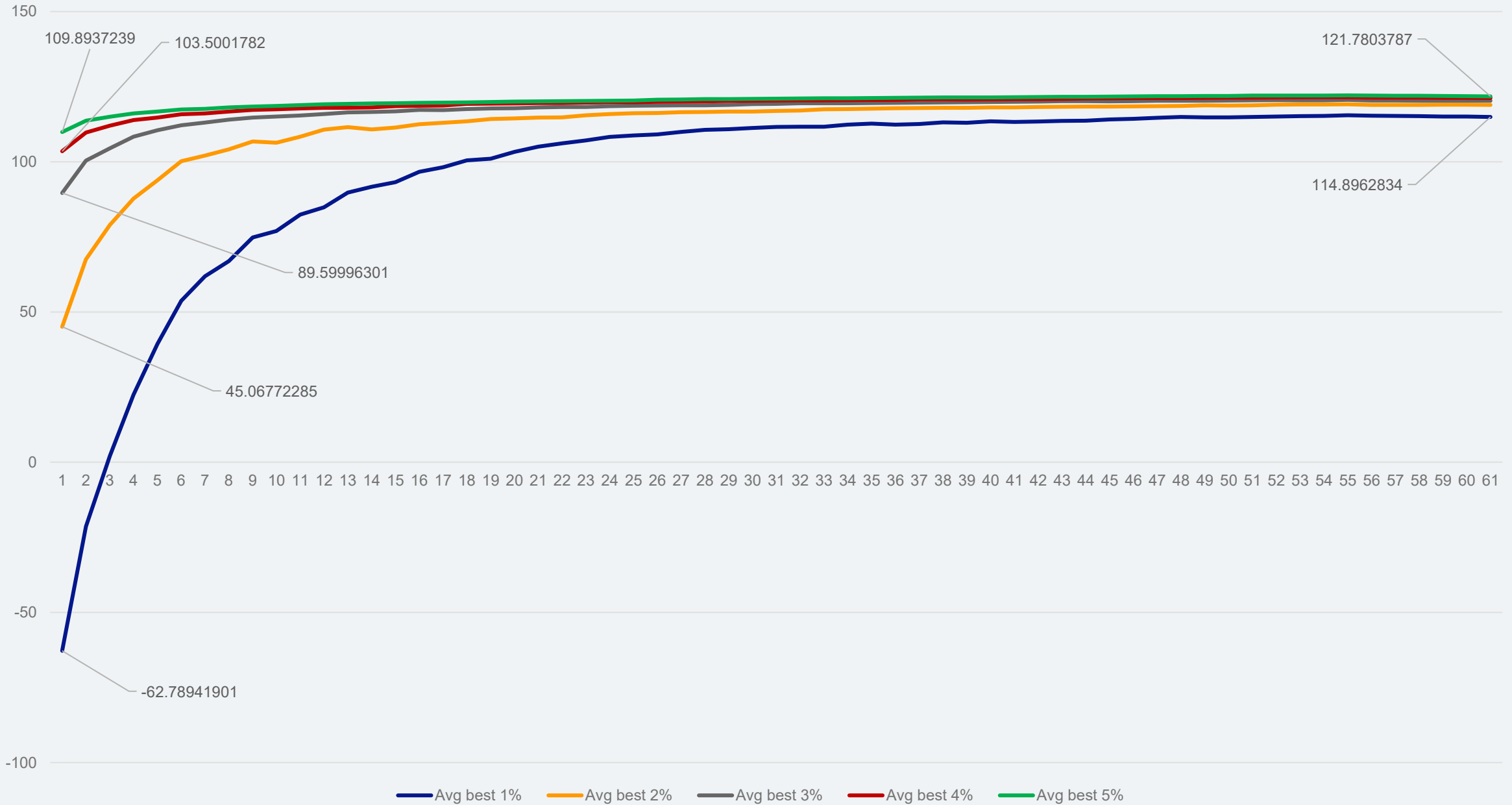
Config.xml

Defines the simulation “rules” via input files and simulation modules.

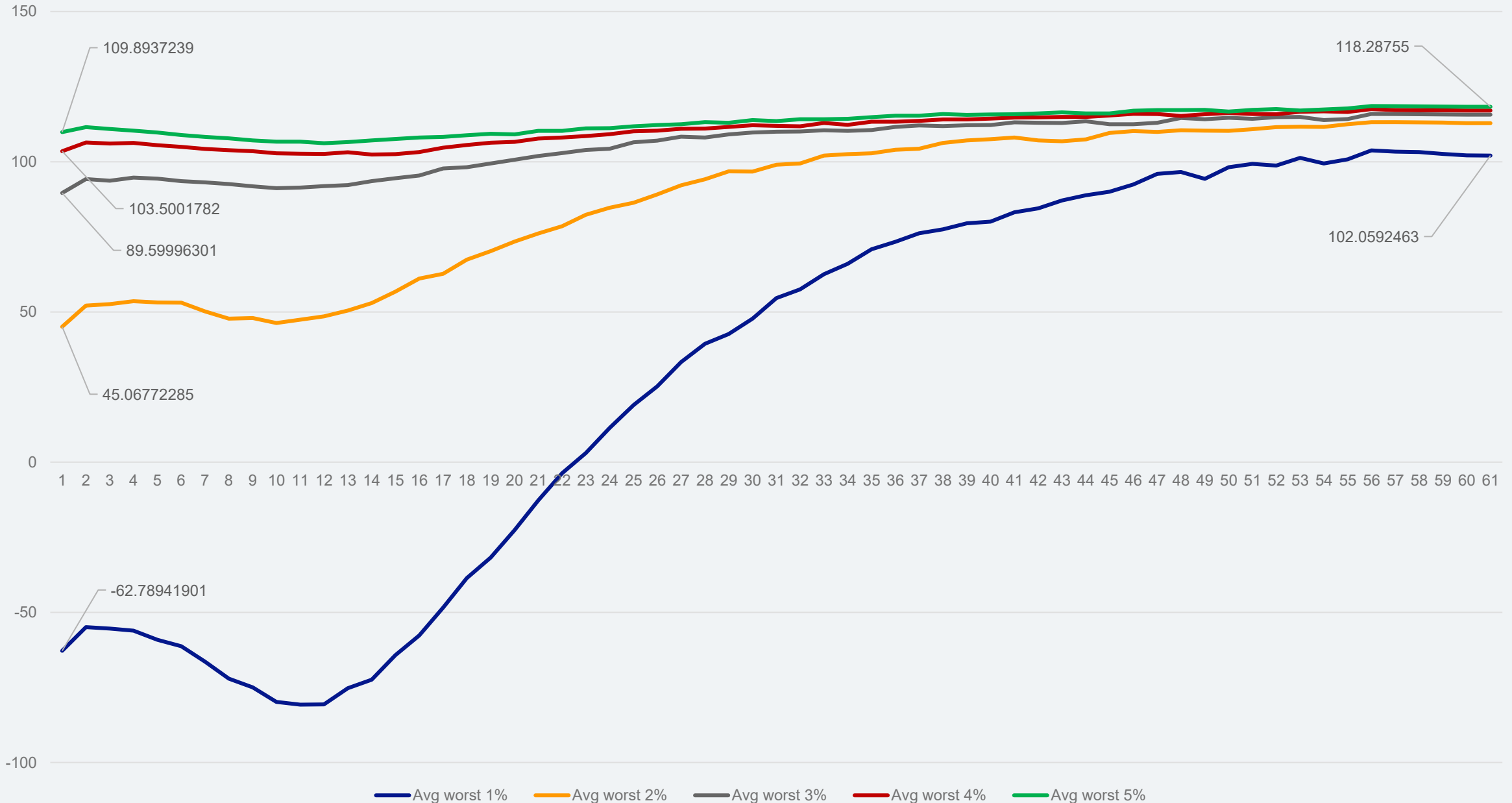
- Controller: The global simulation cycle, set to a total of 60 iterations for this study.
- Scoring: Modified to calibrate on modal split

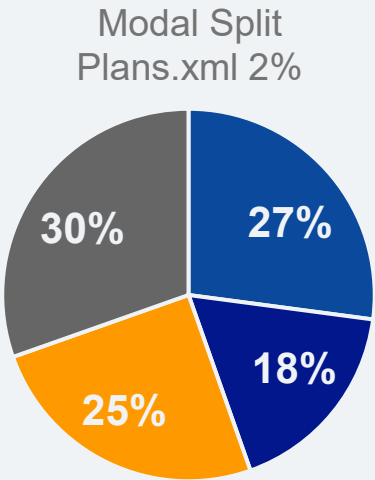
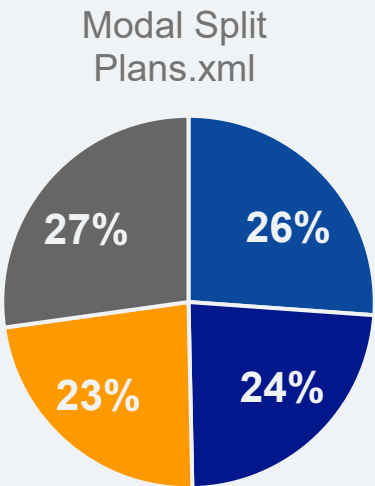
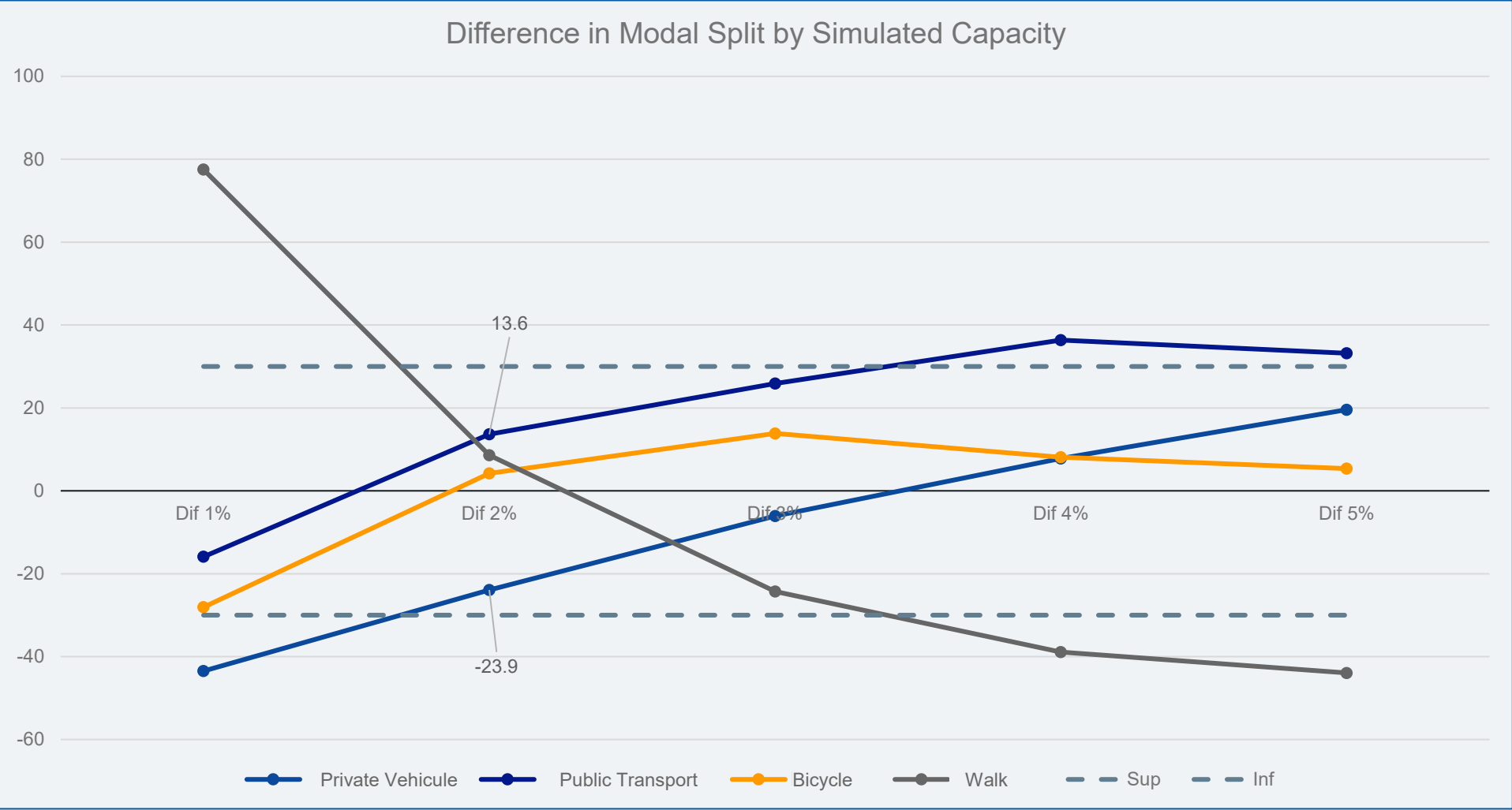


Comparison of Best Utility Curves



Comparison of Worst Utility Curves





Variables / Vehicle Type	Bicycle	Car	Public Transport	Walk
Total Time Traveled (h)	247	165	443	129
Total Distance Traveled (km)	3576	5621	4125	491
Trips	1180	1162	1170	430
Avg. Speed (km/h)	14.49	34.08	9.3	3.8
Avg. distance per trip (km)	3.03	4.84	3.53	1.14

Variables / Vehicle Type	Bicycle	Car	Public Transport	Walk
Total Time Traveled (h)	347	468	511	311
Total Distance Traveled (km)	4116	3921	1942	3662
Trips	1068	986	1200	688
Avg. Speed (km/h)	11.85	8.39	3.8	11.79
Avg. distance per trip (km)	3.85	3.98	1.62	5.32



6.- Results

Results with Progressive Restriction

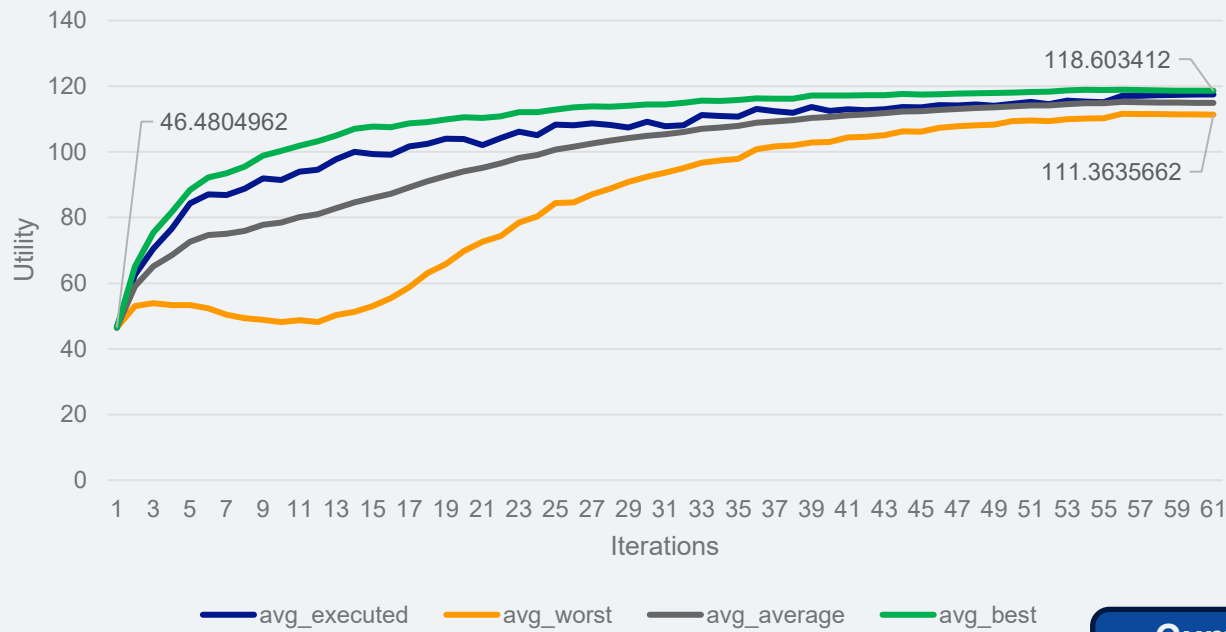
- The calibrated baseline model was used to progressively and cumulatively remove EMT/FGV tram and bus lines.
- Failures were simulated for the entire day (rather than on an hourly basis), assuming that agents had complete information regarding the state of the network.
- Tram 6 – 4
- Bus 98 – 93 – 71 – 40 – 18



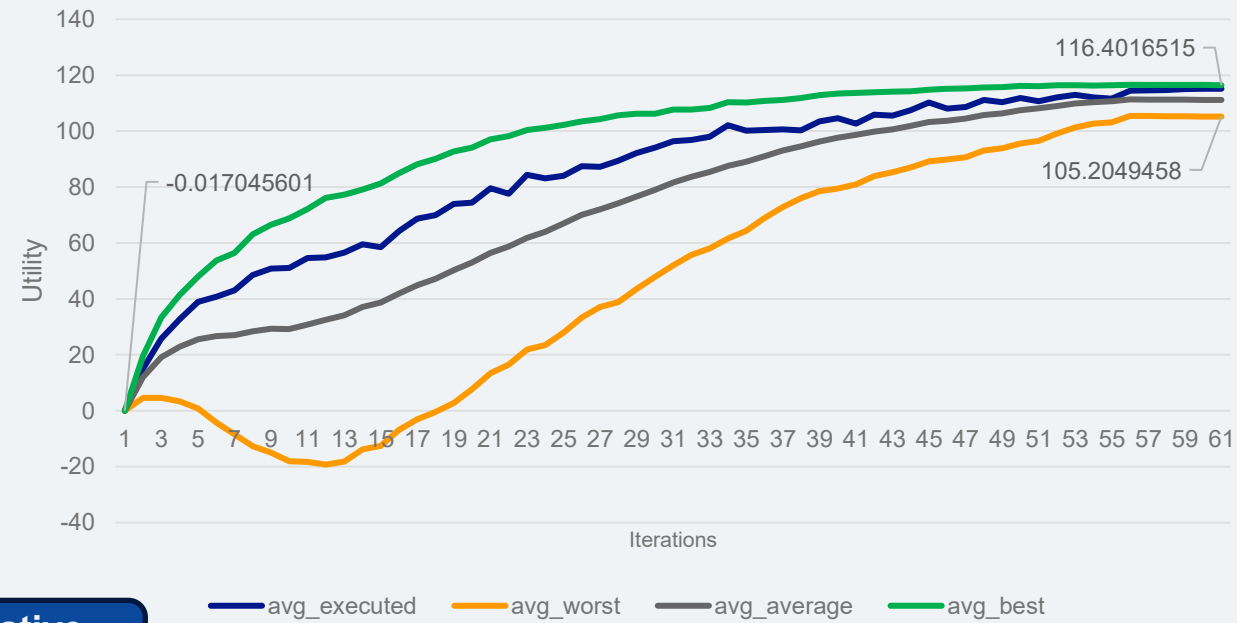
Results with a Single Transport Line

- Each of the 7 lines was simulated operating in isolation to isolate and measure its true demand-capture capacity.
- 

Utility - Without Tram 6 Line



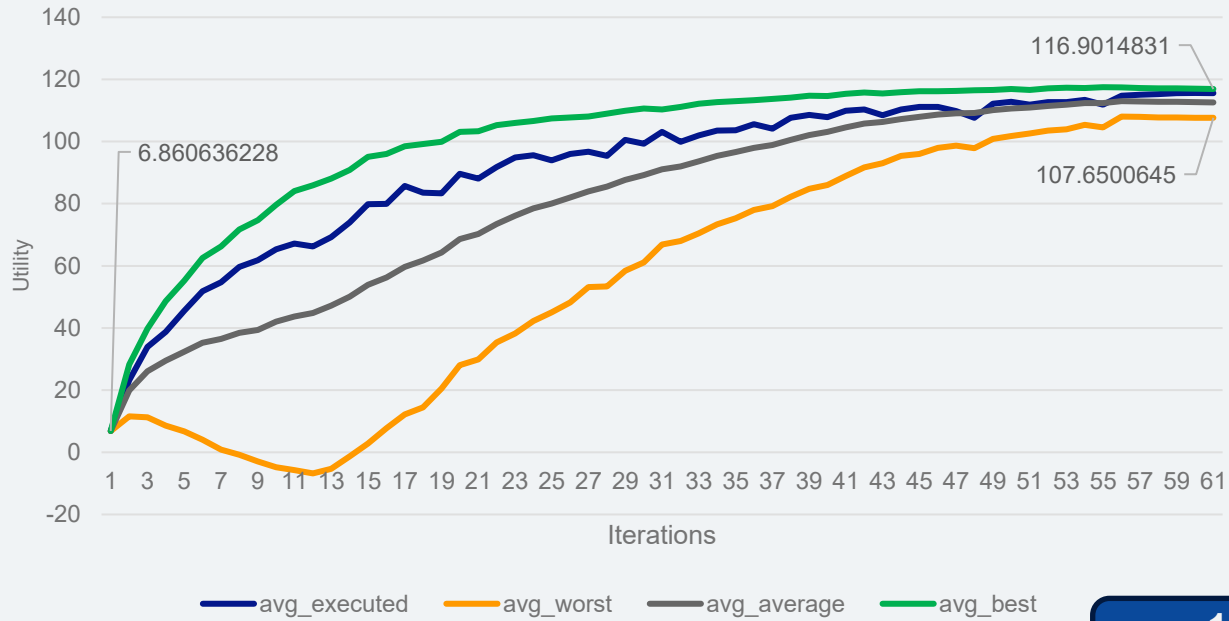
Utility - Without Tram 6/4 and Bus 98/93/71



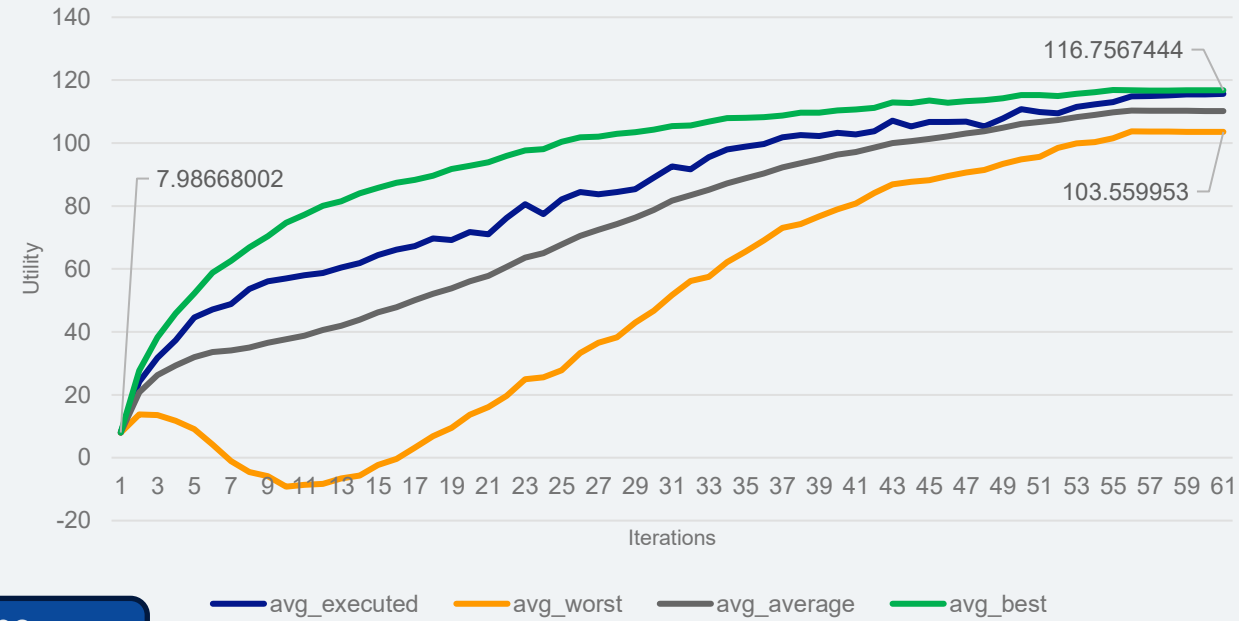
Cumulative Restriction

No simulated Lines	Simulated Lines	Utility Iteration 0 (avg_worst)	Best Utility Iteration 60 (avg_best)	Worst Utility Iteration 60 (avg_worst)	Difference Iteration 60
None	All	46.423	118.873	112.633	5.970
Tram 6	Tram 4, Bus 98, 93, 71, 40 y 18	46.480	118.603	111.363	6.792
Tram 6 y 4	Bus 98, 93, 71, 40 y 18	46.272	118.155	111.783	5.419
Tram 6 y 4, Bus 98	Bus 93, 71, 40 y 18	18.303	117.202	109.198	7.770
Tram 6 y 4, Bus 98, 93	Bus 71, 40 y 18	12.773	116.968	108.486	7.915
Tram 6 y 4, Bus 98, 93, 71	Bus 40 y 18	-0.017	116.401	105.204	11.081

Utility - Only Tram 6 Line

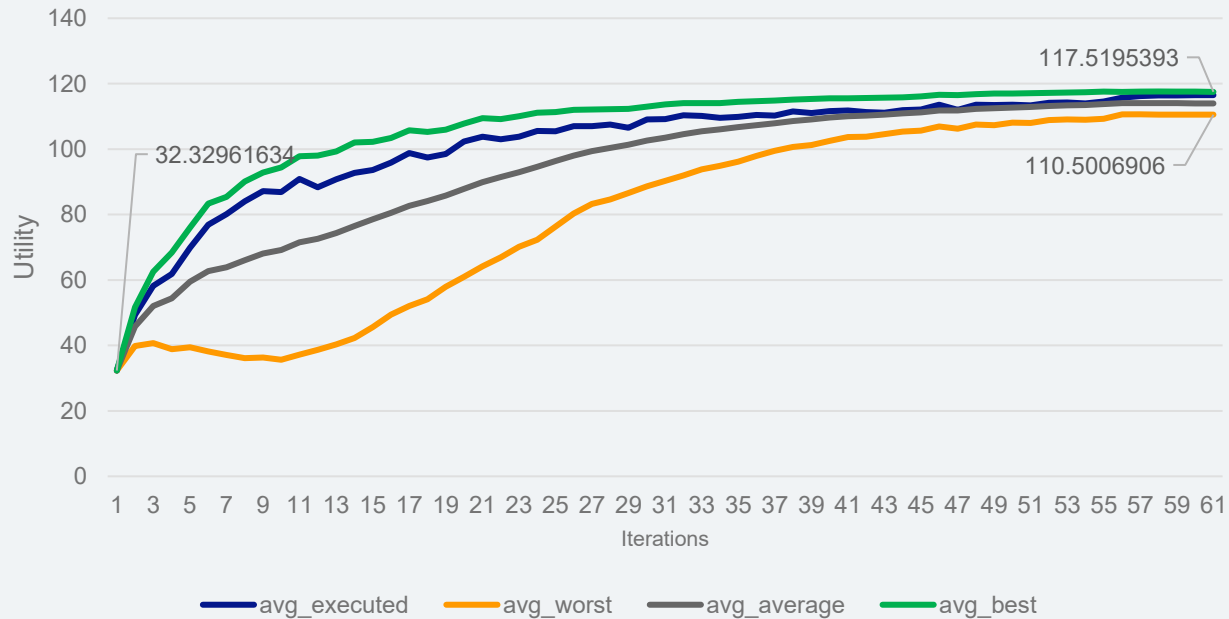


Utility - Only Bus 40 Line



**1 line
Restriction**

Utility - Only Bus 98 Line



Scenario / Utility	Iteration 0 (avg_worst)	Iteration 60 (avg_best)	Iteration 60 (avg_worst)	Iteration 60 Difference
Only Tram 6 Line	6.86	116.901	107.650	9.251
Only Tram 4 Line	-2.567	117.242	103.283	13.959
Only Bus 98 Line	32.329	117.519	110.500	7.019
Only Bus 93 Line	-7.609	116.467	105.330	11.137
Only Bus 71 Line	1.367	116.211	104.347	11.864
Only Bus 40 Line	7.986	116.756	103.559	13.197
Only Bus 18 Line	-6.439	116.285	95.935	20.350

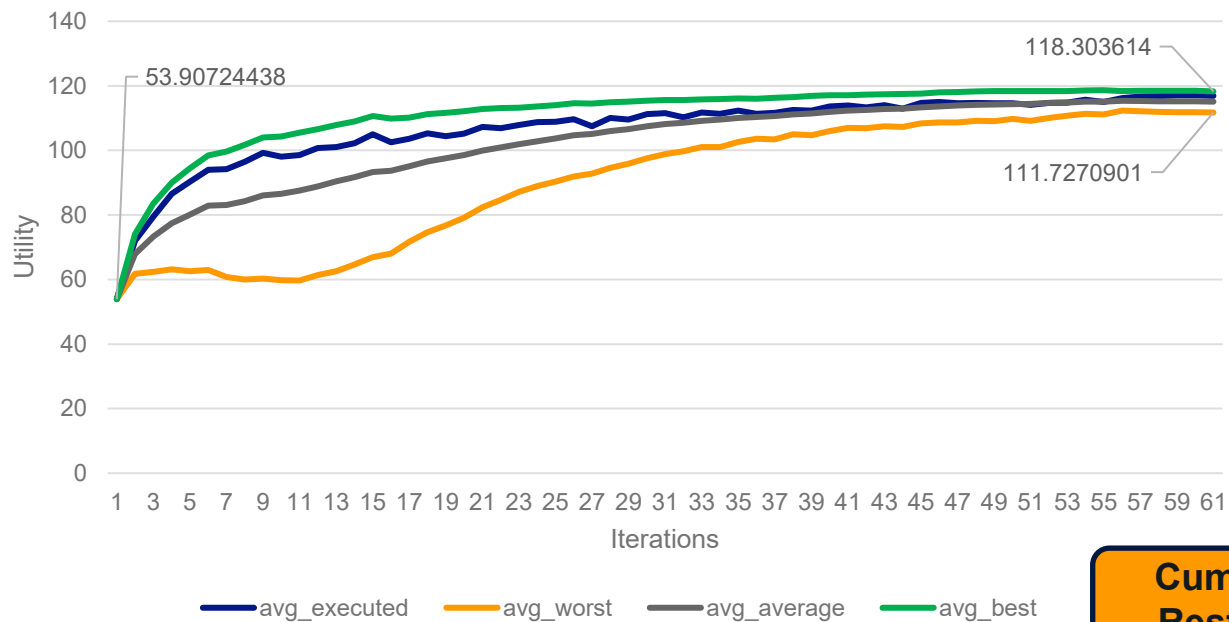


7.- Proposed solutions

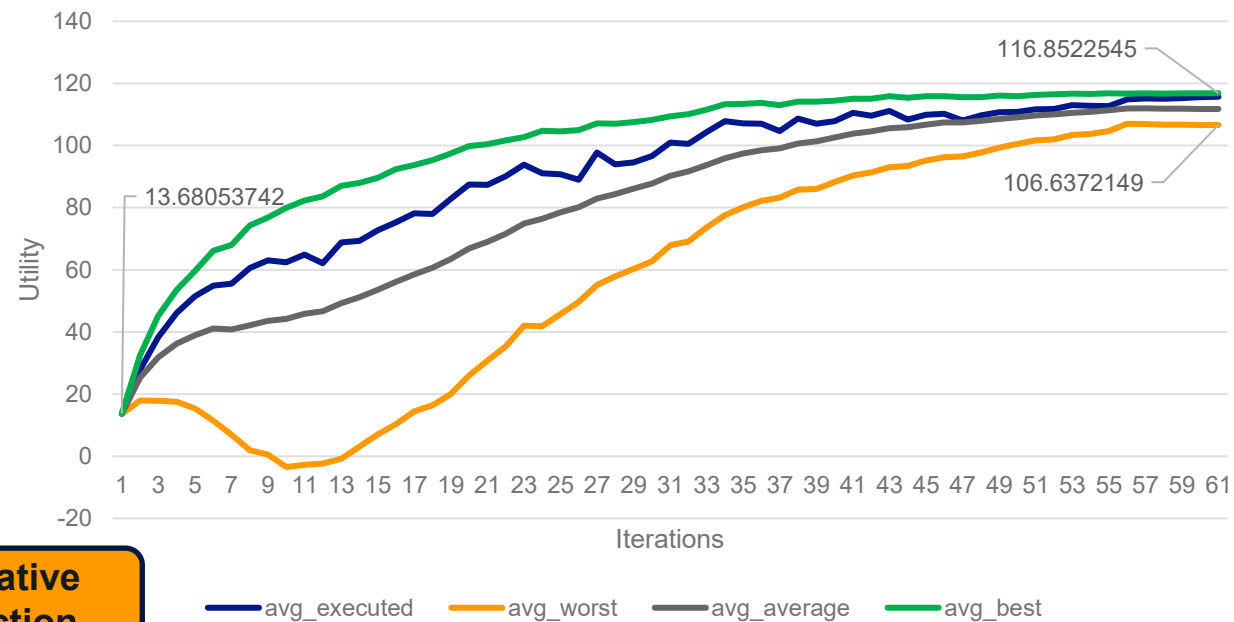
Active Mobility Incentive Solution

- The proposal aims to mitigate transport crises by encouraging bicycle use through economic incentives (discounts or free access to Valenbisi) and the physical expansion of stations at the UPV.
- Daily travel plans were modified using code, based on the assumption that these policies effectively capture 20% of the users affected by public transport disruptions.

Utility - Without Tram 6 Line



Utility - Without Tram 6/4 and Bus 98/93/71 Lines

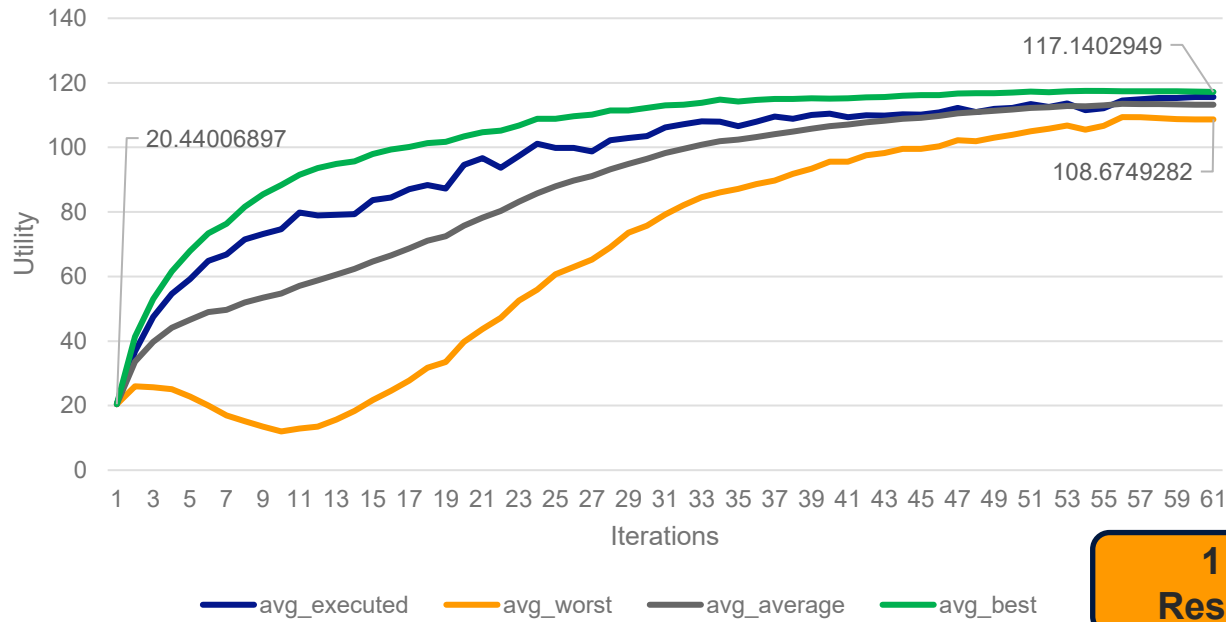


**Cumulative
Restriction**

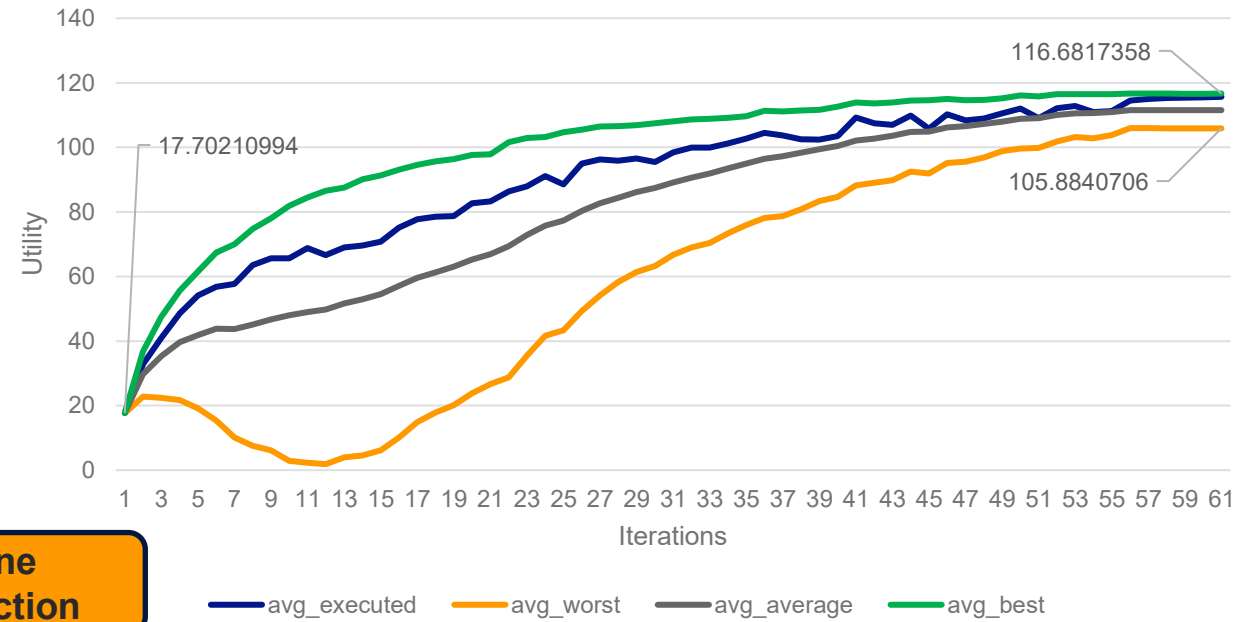
Category / Results	Iteration 0 (avg_worst)	Iteration 60 (avg_best)	Iteration 60 Difference
Mean – Base Scenario	28.37	117.70	111.01
Mean – Solutions	39.65	117.80	110.38
Percentage Difference	39.74%	0.09%	-0.57%

No simulated Lines	Simulated Lines	Utility Iteration 0 (avg_worst)	Best Utility Iteration 60 (avg_best)	Worst Utility Iteration 60 (avg_worst)	Difference Iteration 60
None	All	54.832	118.989	113.290	5.699
Tram 6	Tram 4, Bus 98, 93, 71, 40 y 18	53.907	118.304	111.727	6.577
Tram 6 y 4	Bus 98, 93, 71, 40 y 18	53.348	117.941	111.751	6.191
Tram 6 y 4, Bus 98	Bus 93, 71, 40 y 18	34.142	117.360	109.136	8.224
Tram 6 y 4, Bus 98, 93	Bus 71, 40 y 18	27.981	117.397	109.786	7.611
Tram 6 y 4, Bus 98, 93, 71	Bus 40 y 18	13.681	116.852	106.637	10.215

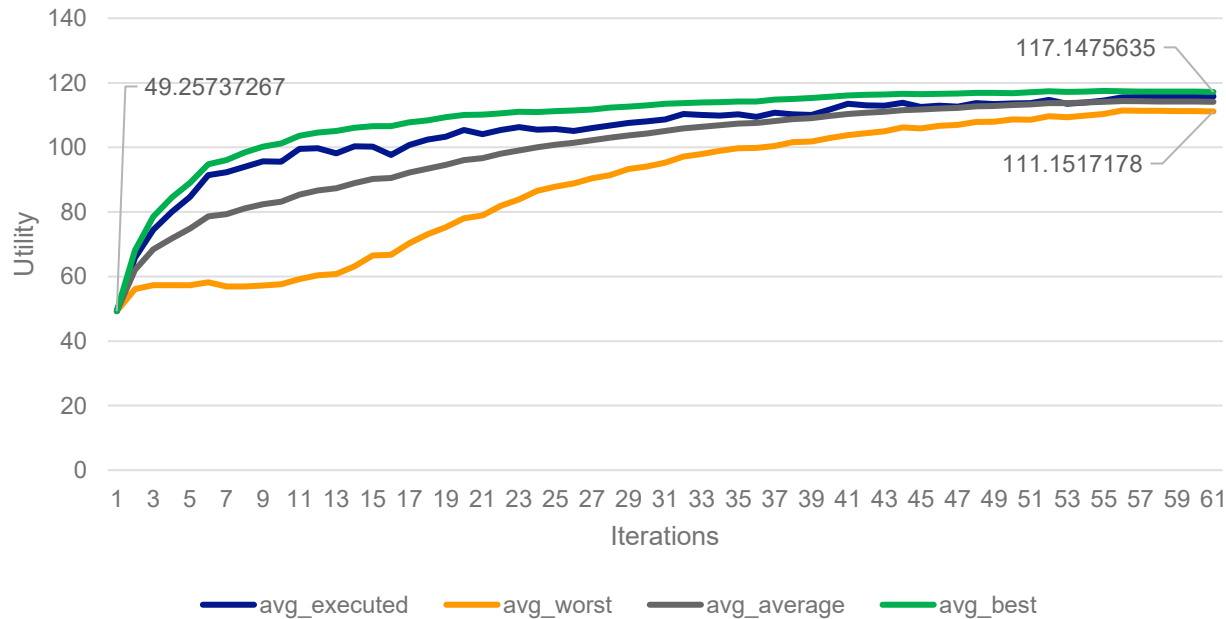
Utilidades - Última Línea Tranvía 6



Utilidades - Última Línea Bus 40

**1 Line
Restriction**

Utilidades - Última Línea Bus 98



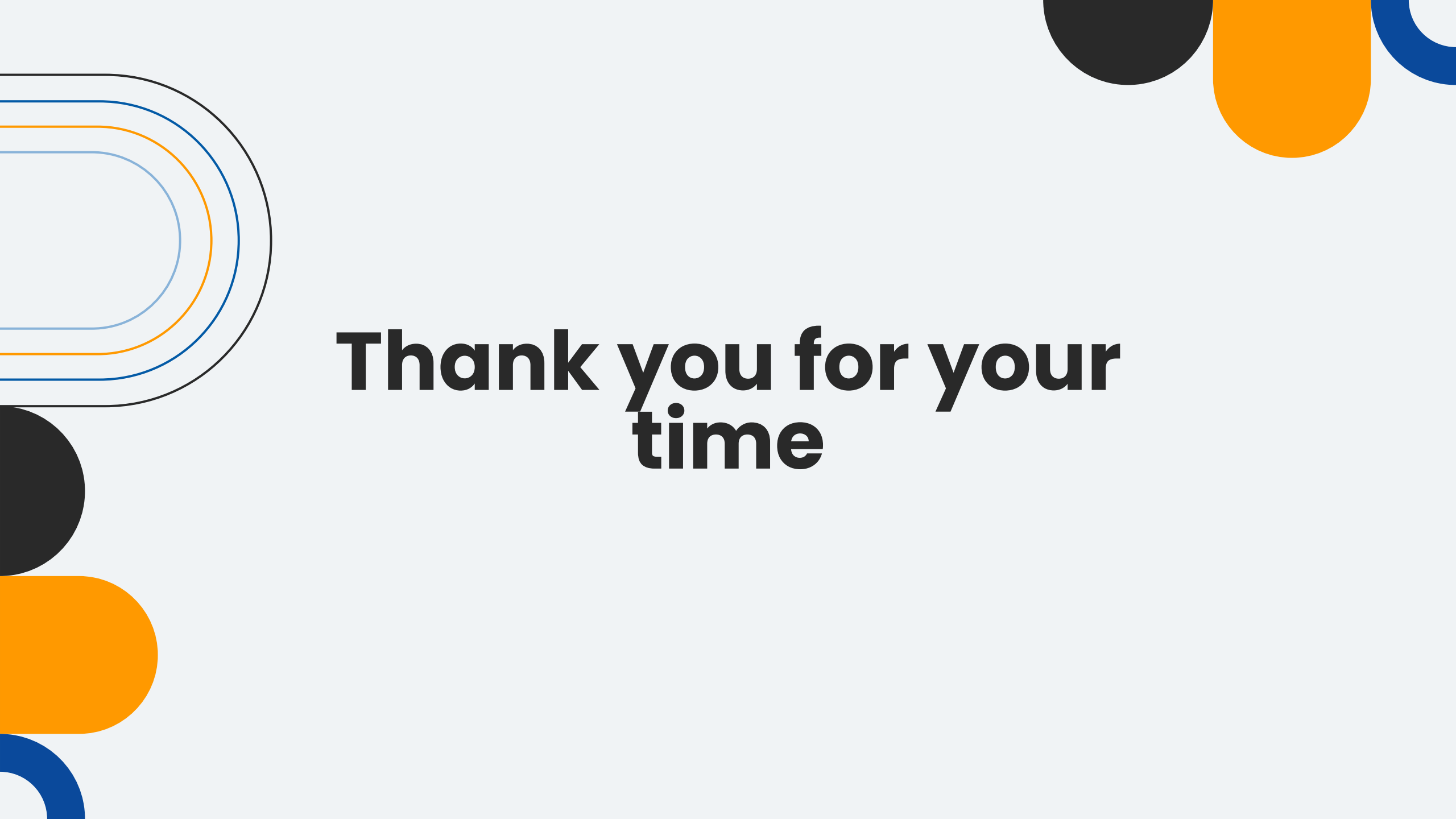
Category / Results	Iteration 0 (avg_worst)	Iteration 60 (avg_best)	Iteration 60 Difference
Mean – Base Scenario	4.561	116.769	104.373
Mean – Solutions	17.736	116.994	107.547
Percentage Difference	288.86%	0.19%	3.04%

Scenario / Utility	Iteración 0 (avg_worst)	Iteration 60 (avg_best)	Iteration 60 (avg_worst)	Iteration 60 Difference
Only Tram 6 Line	20.440	117.140	108.675	8.465
Only Tram 4 Line	12.129	117.734	107.833	9.901
Only Bus 98 Line	49.257	117.148	111.152	5.996
Only Bus 93 Line	7.750	116.474	107.642	8.833
Only Bus 71 Line	12.430	116.874	108.375	8.499
Only Bus 40 Line	17.702	116.682	105.884	10.798
Only Bus 18 Line	4.445	116.906	103.268	13.638



8.- Conclusions

- Agent-Based Models (MATSim) have proven highly effective for assessing dynamic mobility, with the acquisition of disaggregated data representing the only major challenge to their development.
 - Valencia's public transport system features significant redundancy; thanks to overlapping routes, isolated failures have far less impact on mobility than might be expected.
 - The EMT's bus line 98 and the FGV's tram line 6 stand out as the absolute pillars of campus mobility, capturing the highest volume of passenger demand.
 - Encouraging active mobility helps mitigate the immediate impact of a crisis, yet its effectiveness is limited by the diversity of users and the distances they need to travel. While it is an excellent solution for short and medium distances, it is insufficient on its own to absorb the collapse of the transport network.
- 

The image features a light blue background with several decorative graphic elements. In the top right corner, there are three overlapping rounded shapes: a dark grey one, an orange one, and a blue one. In the bottom left corner, there are three overlapping rounded shapes: a dark grey one, an orange one, and a blue one. On the left side, there is a series of four concentric, rounded rectangular outlines in black, blue, orange, and light blue. The text "Thank you for your time" is centered in the middle of the image in a bold, dark grey sans-serif font.

**Thank you for your
time**



Appendices

Codes

```
<module name=qsim>
  <param name=mainMode value=car,bicycle />
  <param name=trafficDynamics value=kinematicWaves />
  <param name=startTime value=00:00:00 />
  <param name=endTime value=36:00:00 />
  <param name=vehiclesSource value=modeVehicleTypesFromVehiclesData />
  <param name=snapshotperiod value = 00:00:00/>
  <param name=flowCapacityFactor value=1 />
  <param name=storageCapacityFactor value=1/>
</module>
```

Parámetros	Modos			
	Transporte Público	Bicicleta	Coche	Caminata
Constant	-0.7	-1.1	1.0	0.0
marginalUtilityOfDistance_util_m	0.0	-0.0005	0.0	0.0
marginalUtilityOfTraveling_util_hr	-6.0	-6.0	-6.0	-6.0
monetaryDistanceRate	0.0	0.0	-0.0001	0.0

```
<module name=scoring>
  <param name=learningRate value=1.0 />
  <param name=brainExpBeta value=2.0 />
  <param name=lateArrival value=-18 />
  <param name=earlyDeparture value=-0 />
  <param name=performing value=+6 />
  <param name=waiting value=-0 />

  <parameterset type=modeParams>
    <param name=mode value=pt/>
    <param name=constant value=-0.7 />
    <param name=marginalUtilityOfDistance_util_m value=0.0 />
    <param name=marginalUtilityOfTraveling_util_hr value=-6.0 />
    <param name=monetaryDistanceRate value=0.0 />
  </parameterset>

  <parameterset type=activityParams>
    <param name=activityType value=s />
    <param name=priority value=1 />
    <param name=typicalDuration value=05:00:00 />
    <param name=minimalDuration value=01:00:00 />
  </parameterset>
</module>
```

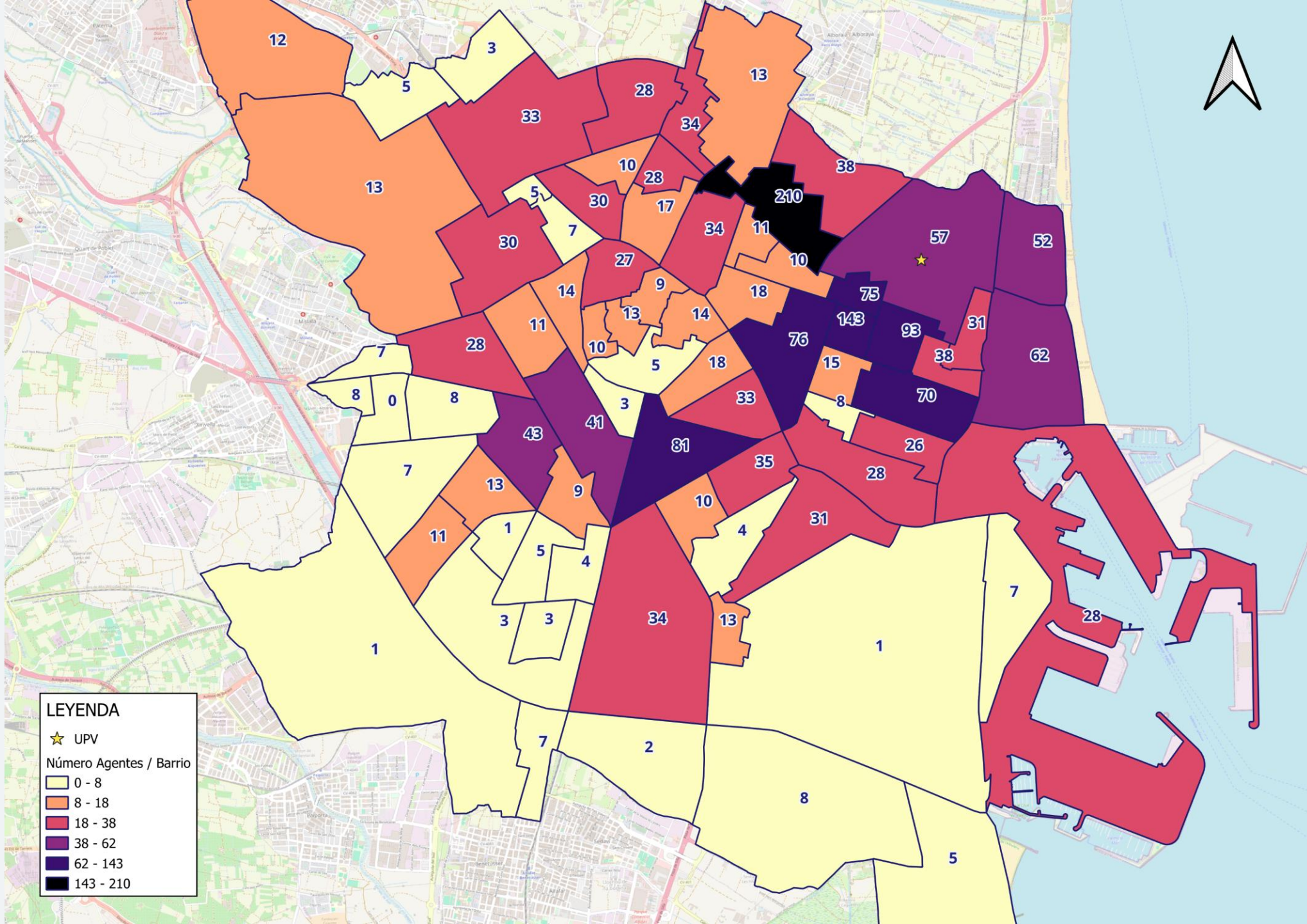

Codes

```
<module name=transitRouter>
  <param name=searchRadius value=250 />
  <param name=extensionRadius value=500 />
  <param name=maxBeelineWalkConnectionDistance value=200.0 />
  <param name=additionalTransferTime value=180.0 />
  <param name=directWalkFactor value=1.0 />
</module>
```

```
<module name=routing>
  <param name=networkModes value=car,bicycle />
  <param name=clearDefaultTeleportedModeParams value=true />
  <parameterset type=teleportedModeParameters>
    <param name=mode value=walk/>
    <param name=beelineDistanceFactor value=1.3/>
    <param name=teleportedModeSpeed value=1.0555556/>
  </parameterset>
</module>
```

```
<module name=replanning>
  <param name=fractionOfIterationsToDisableInnovation value=0.7/>
  <parameterset type=strategysettings>
    <param name=strategyName value=ChangeExpBeta/>
    <param name=weight value=0.7/>
  </parameterset>
  <parameterset type=strategysettings>
    <param name=strategyName value=ReRoute/>
    <param name=weight value=0.10/>
  </parameterset>
  <parameterset type=strategysettings>
    <param name=strategyName value=SubtourModeChoice/>
    <param name=weight value=0.10/>
  </parameterset>
  <param name=strategyName value=TimeAllocationMutator/>
  <param name=weight value=0.10/>
</parameterset>
</module>

<module name=subtourModeChoice>
  <param name=considerCarAvailability value=true />
  <param name=modes value=car,pt,bicycle,walk />
  <param name=chainBasedModes value=car,bicycle />
</module>
```



LEYENDA

★ UPV

Número Agentes / Barrio

0 - 8

8 - 18

18 - 38

38 - 62

62 - 143

143 - 210

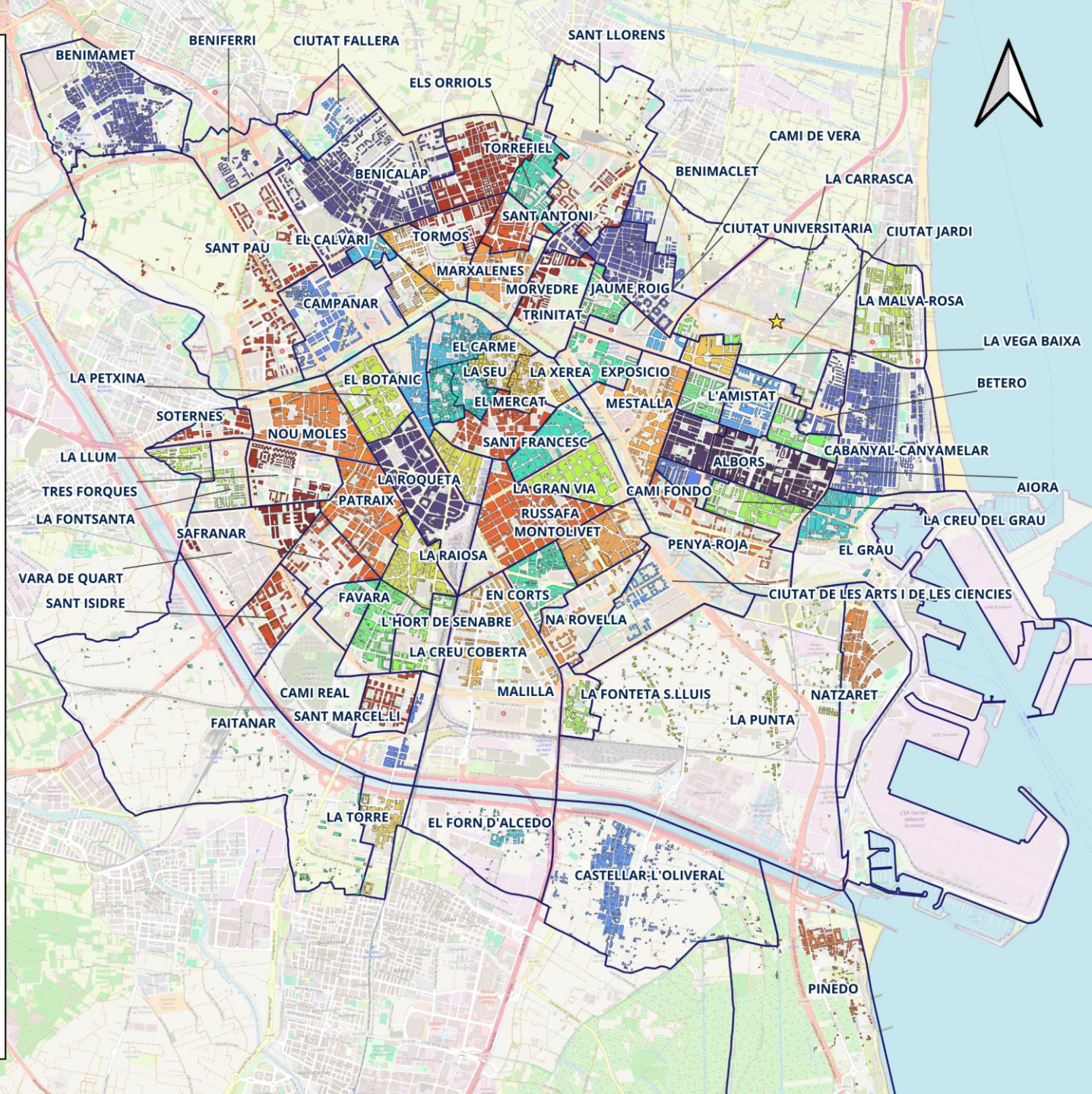
★ UPV



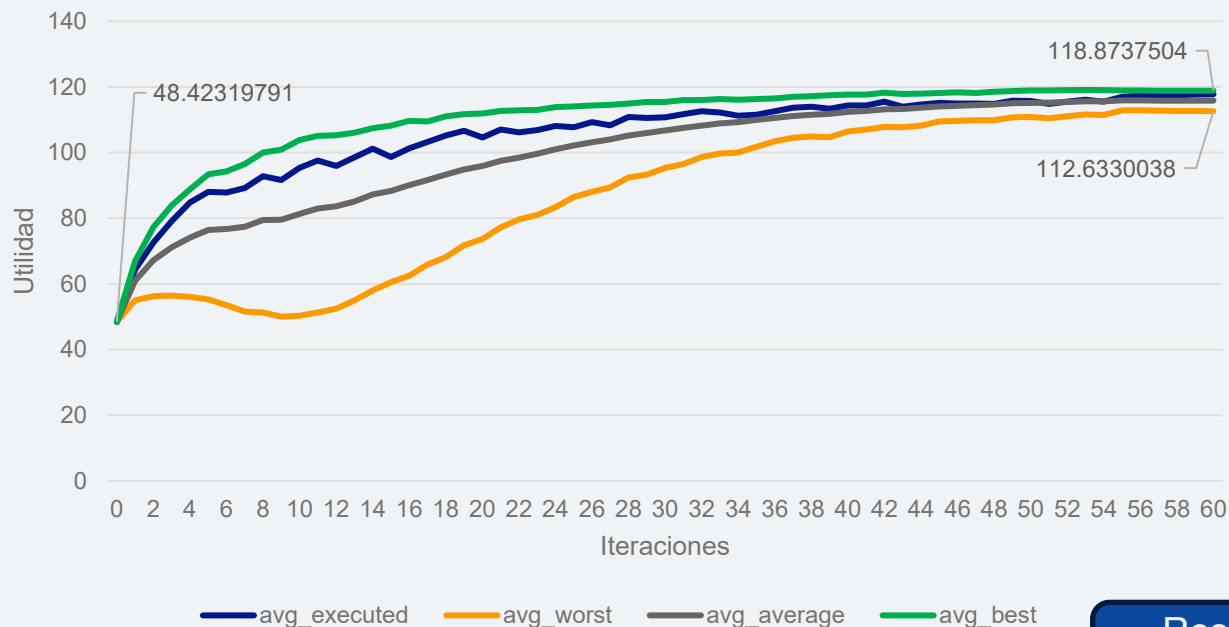
Intersección

- | |
|---|
| AIORA |
| ALBORS |
| ARRANCAPINS |
| BENICALAP |
| BENIFERRI |
| BENIMACLET |
| BENIMAMET |
| BETERO |
| CABANYAL-CANYAMELAR |
| CAMI DE VERA |
| CAMI FONDO |
| CAMI REAL |
| CAMPANAR |
| CASTELLAR-L'OLIVERAL |
| CIUTAT DE LES ARTS I
DE LES CIENCIES |
| CIUTAT FALLERA |
| CIUTAT JARDI |
| CIUTAT UNIVERSITARIA |
| EL BOTANIC |
| EL CALVARI |
| EL CARME |
| EL FORN D'ALCEDO |
| EL GRAU |
| EL MERCAT |
| EL PILAR |
| EL PLA DEL REMEI |
| ELS ORRIOLS |
| EN CORTS |
| EXPOSICIO |
| FAITANAR |
| FAVARA |
| JAUME ROIG |
| L'AMISTAT |
| L'HORT DE SENABRE |
| L'ILLA PERDUDA |
| LA CARRASCA |
| LA CREU COBERTA |
| LA CREU DEL GRAU |

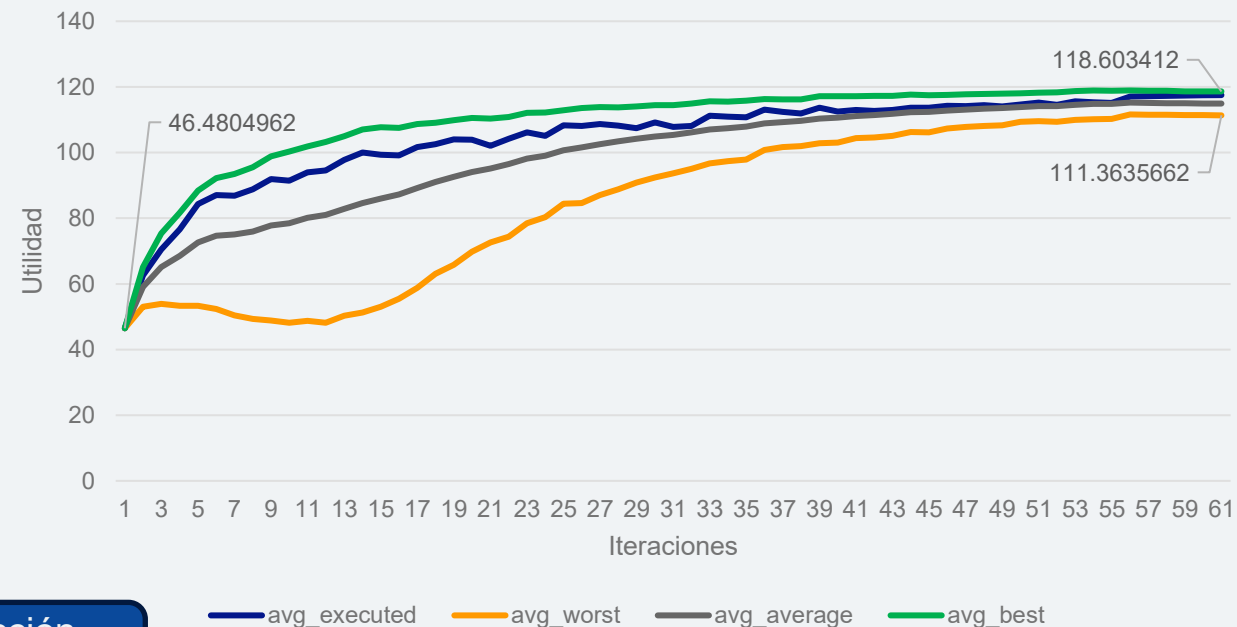
- | |
|--------------------|
| LA FONTETA S.LLUIS |
| LA FONTSANTA |
| LA GRAN VIA |
| LA LLUM |
| LA MALVA-ROSA |
| LA PETXINA |
| LA PUNTA |
| LA RAIOSA |
| LA ROQUETA |
| LA SEU |
| LA TORRE |
| LA VEGA BAIXA |
| LA XEREA |
| LES TENDETES |
| MALILLA |
| MARXALENES |
| MESTALLA |
| MONTOLIVET |
| MORVEDRE |
| NA ROVELLA |
| NATZARET |
| NOU MOLES |
| PATRAIX |
| PENYA-ROJA |
| PINEDO |
| RUSSAFA |
| SAFRANAR |
| SANT ANTONI |
| SANT FRANCESC |
| SANT ISIDRE |
| SANT LLORENS |
| SANT MARCEL.LI |
| SANT PAU |
| SOTERNES |
| TORMOS |
| TORREFIEL |
| TRES FORQUES |
| TRINITAT |
| VARA DE QUART |



Utilidades - Totalidad de la oferta de Transporte

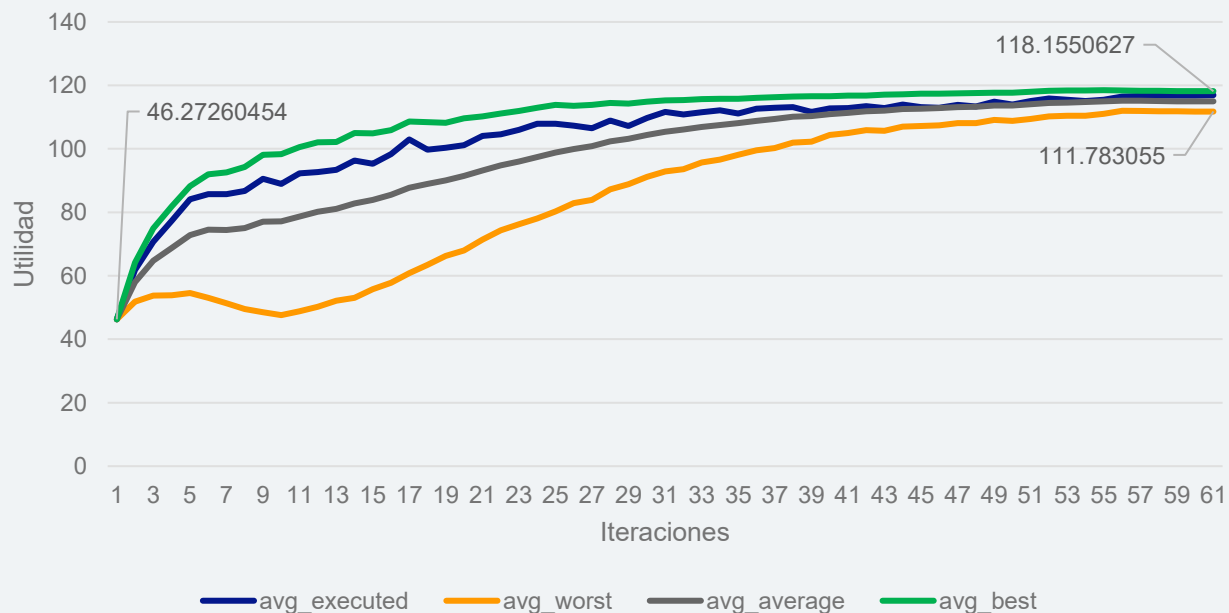


Utilidades - sin Línea Tranvía 6

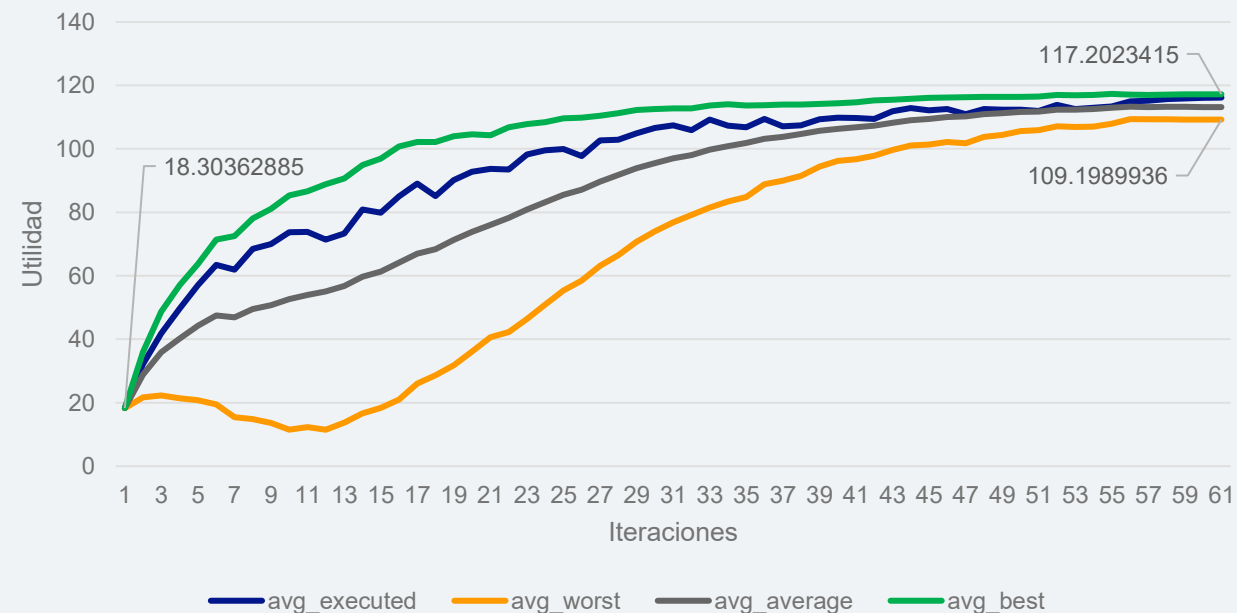


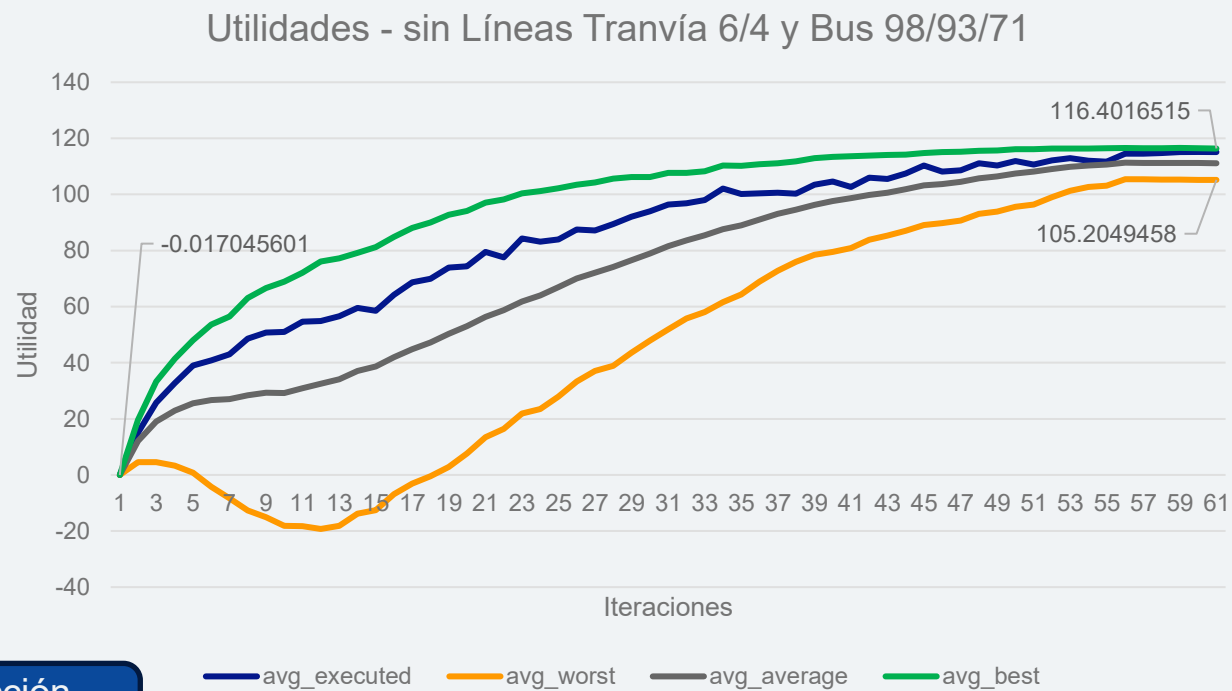
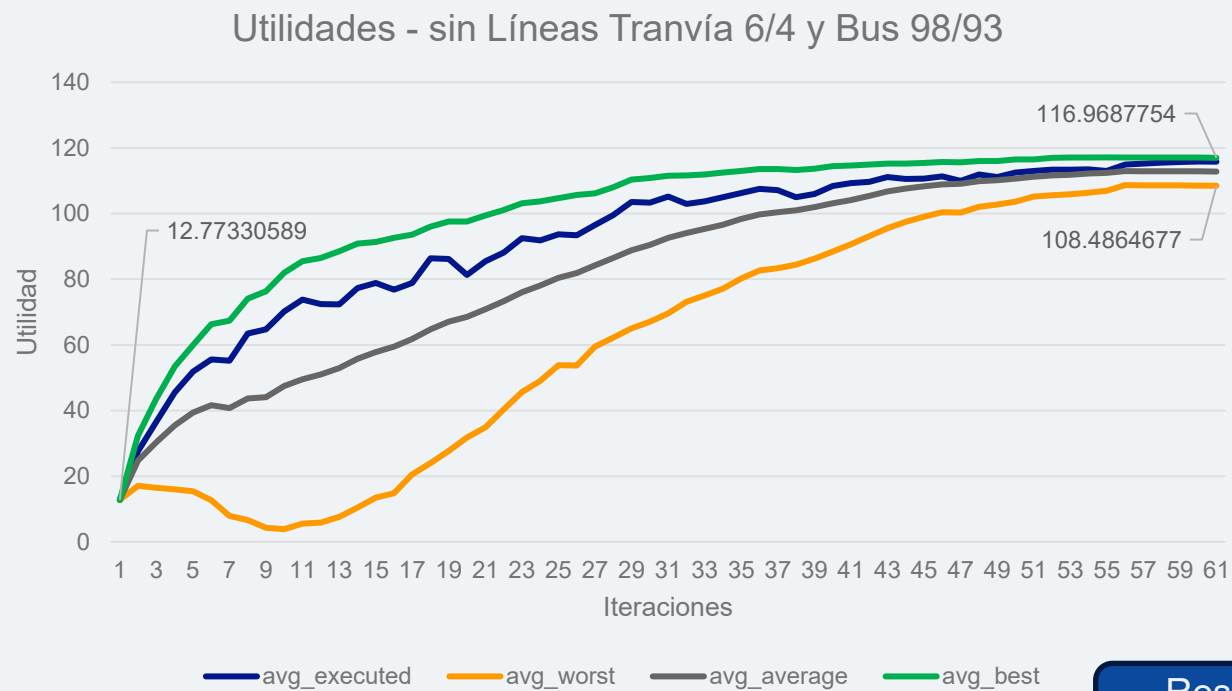
Restricción Acumulada

Utilidades - sin Línea Tranvía 6/4



Utilidades - sin Líneas Tranvía 6/4 y Bus 98

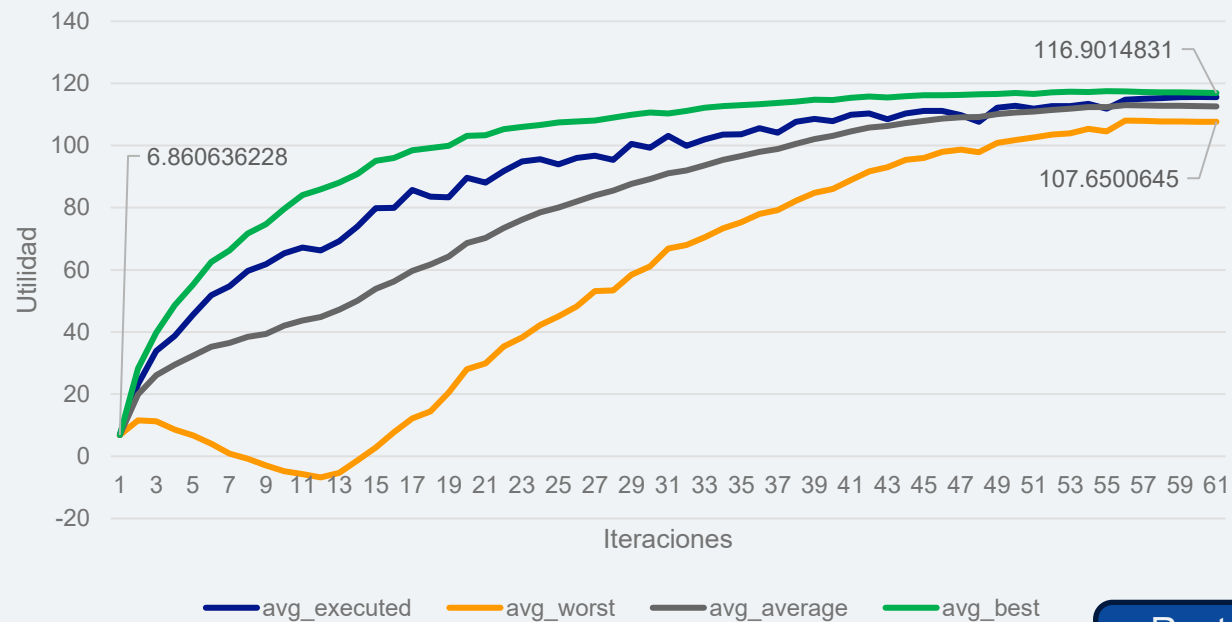




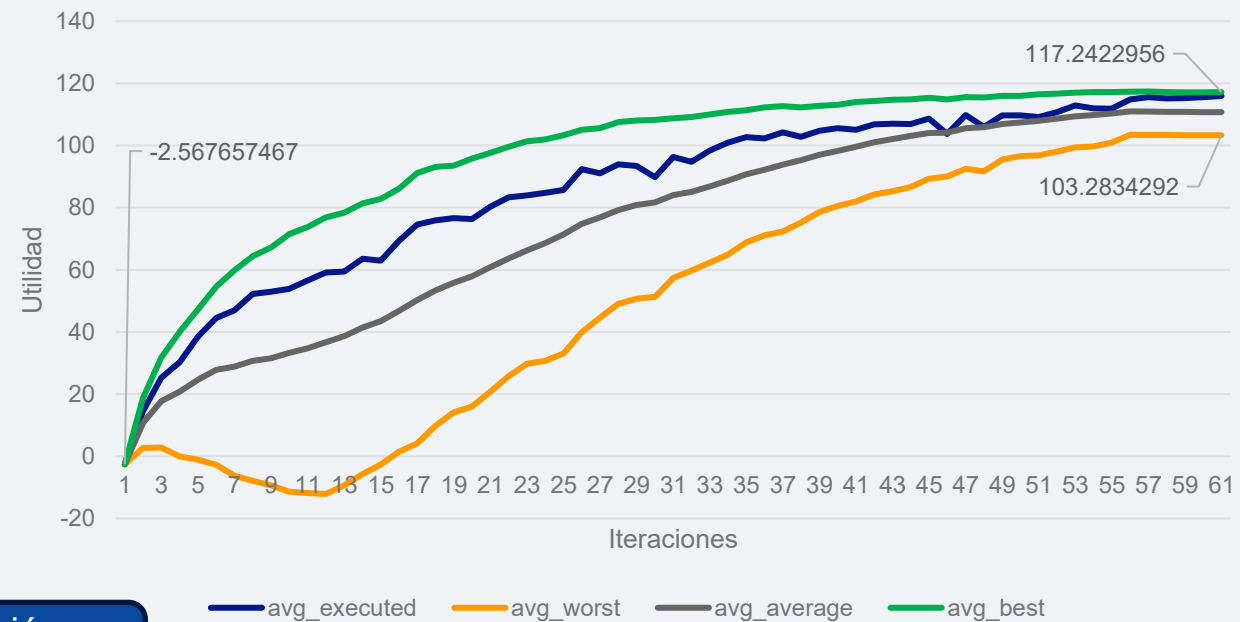
Restricción
Acumulada

Líneas No simuladas	Líneas Simuladas	Utilidad Iteración 0 (avg_worst)	Mejor Utilidad Iteración 60 (avg_best)	Peor Utilidad Iteración 60 (avg_worst)	Diferencia Iteración 60
Ninguna	Todas	46.423	118.873	112.633	5.970
Tranvía 6	Tranvía 4, Bus 98, 93, 71, 40 y 18	46.480	118.603	111.363	6.792
Tranvía 6 y 4	Bus 98, 93, 71, 40 y 18	46.272	118.155	111.783	5.419
Tranvía 6 y 4, Bus 98	Bus 93, 71, 40 y 18	18.303	117.202	109.198	7.770
Tranvía 6 y 4, Bus 98, 93	Bus 71, 40 y 18	12.773	116.968	108.486	7.915
Tranvía 6 y 4, Bus 98, 93, 71	Bus 40 y 18	-0.017	116.401	105.204	11.081

Utilidades - Última Línea Tranvía 6

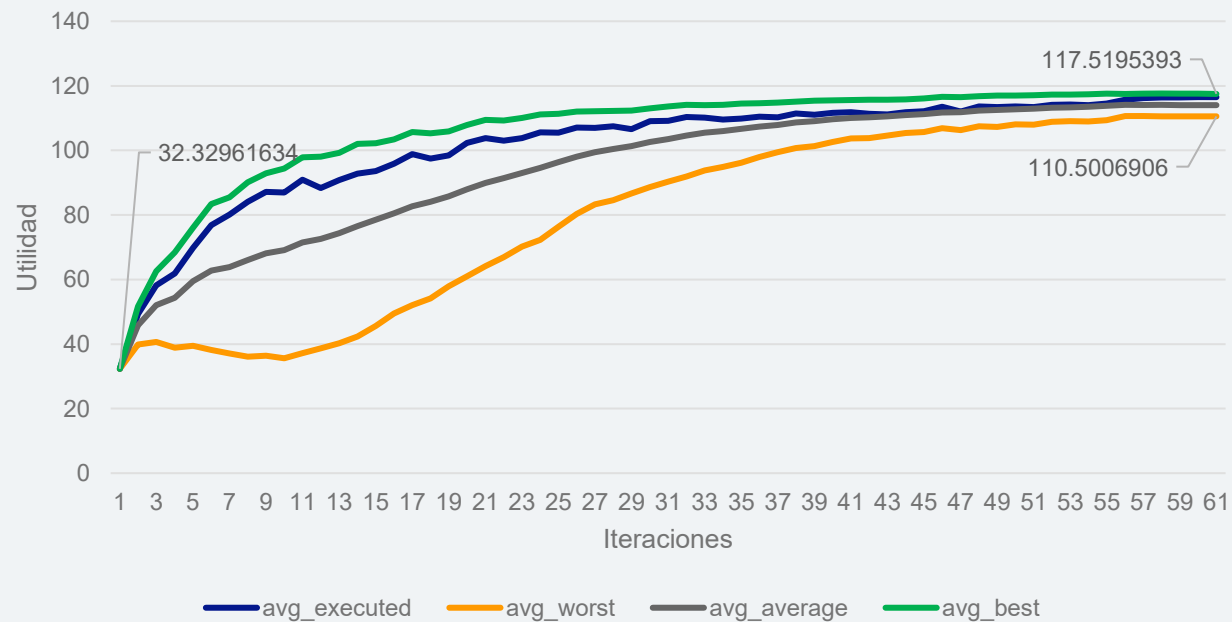


Utilidades - Última Línea Tranvía 4

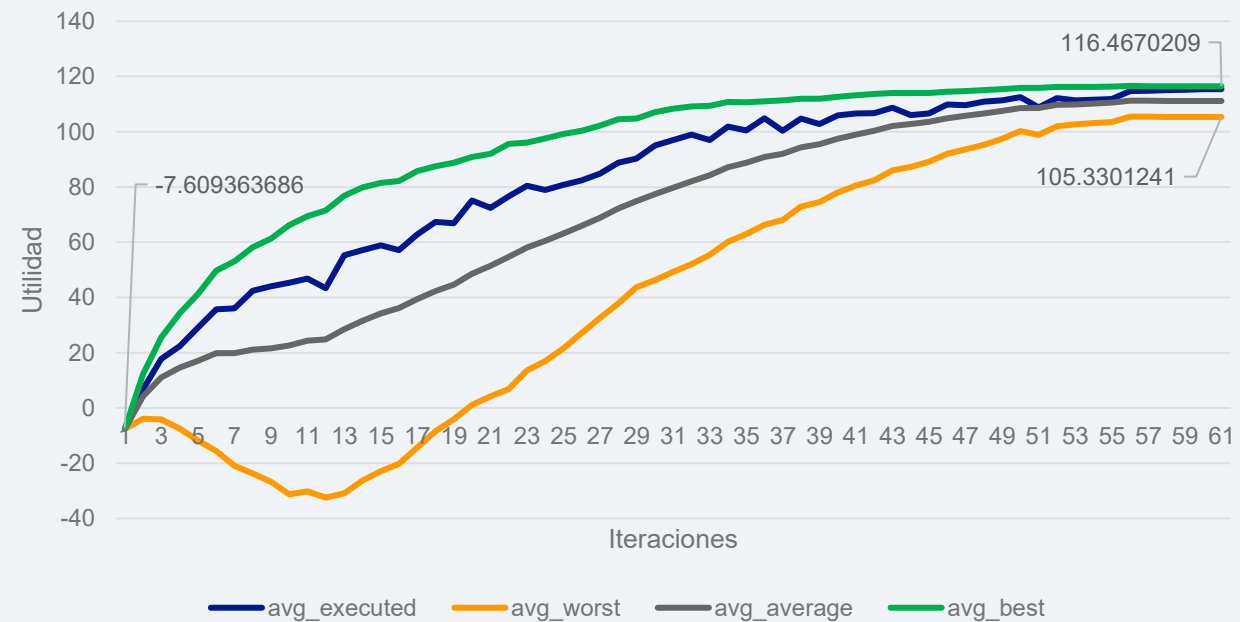


Restricción a una línea

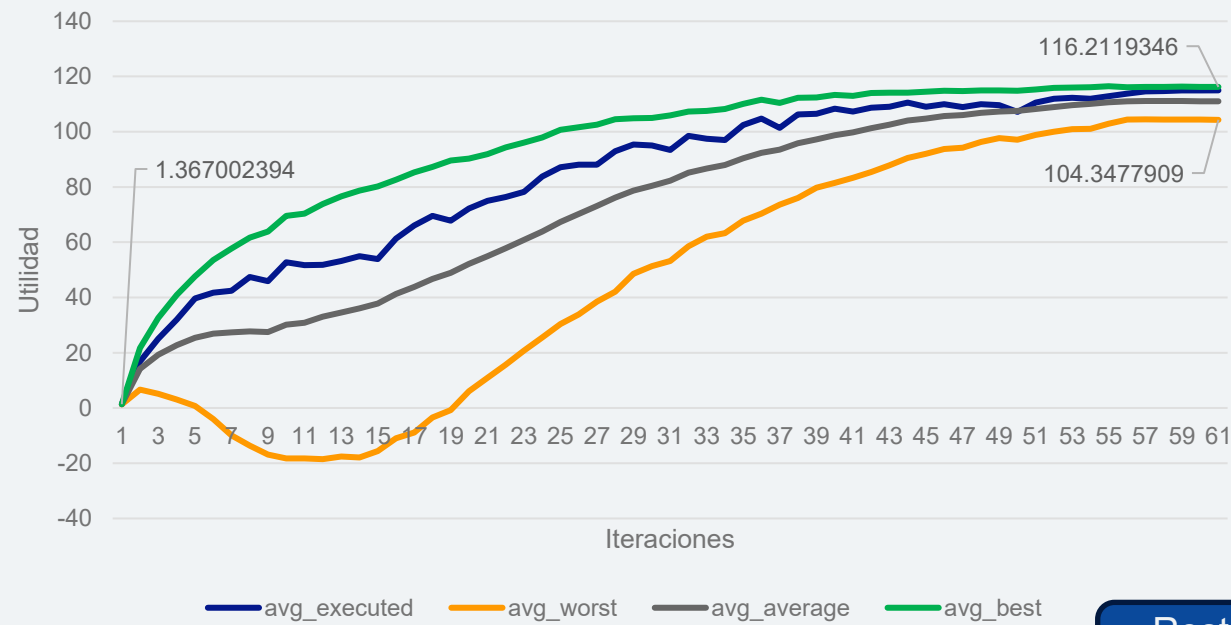
Utilidades - Última Línea Bus 98



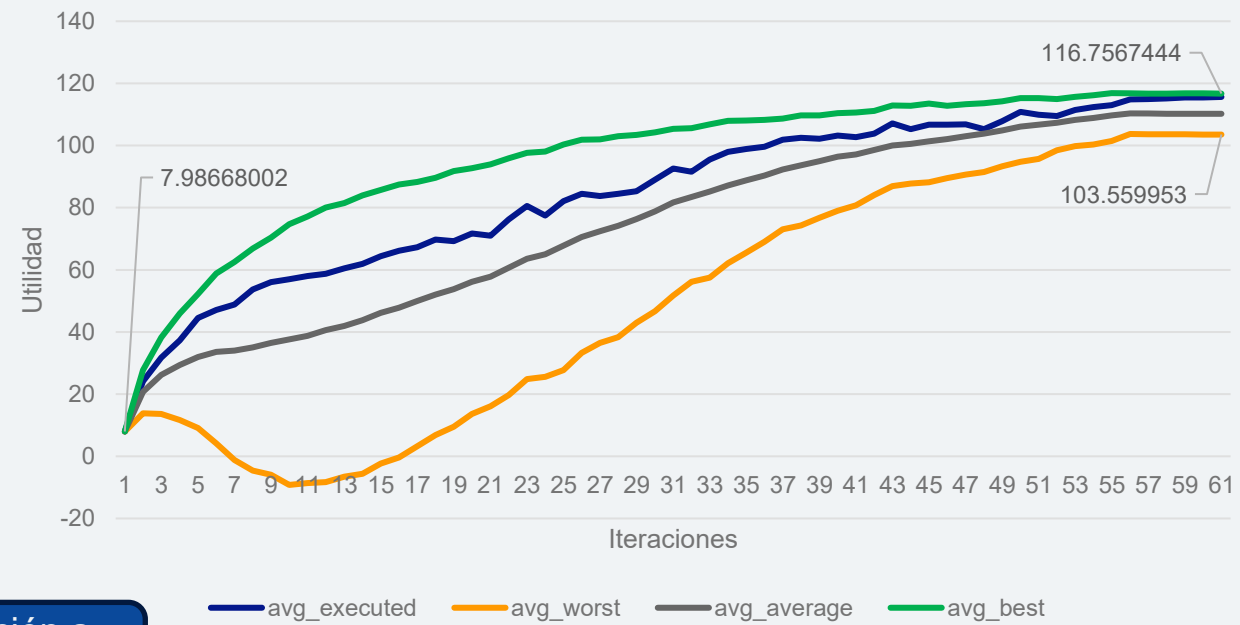
Utilidades - Última Línea Bus 93



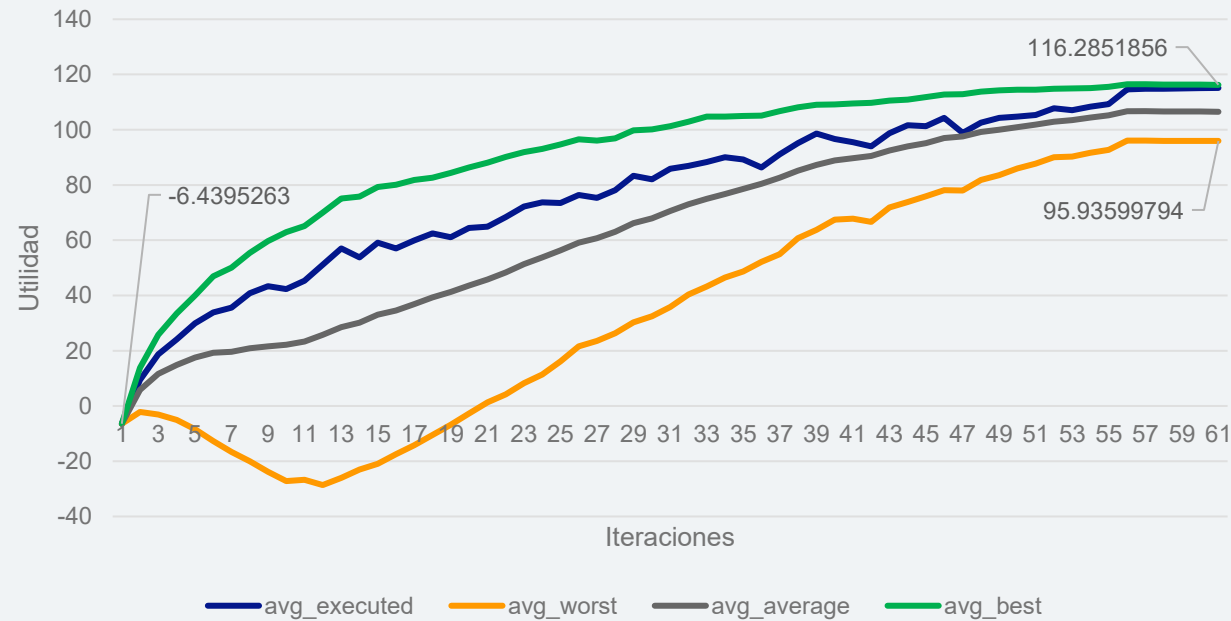
Utilidades - Última Línea Bus 71



Utilidades - Última Línea Bus 40



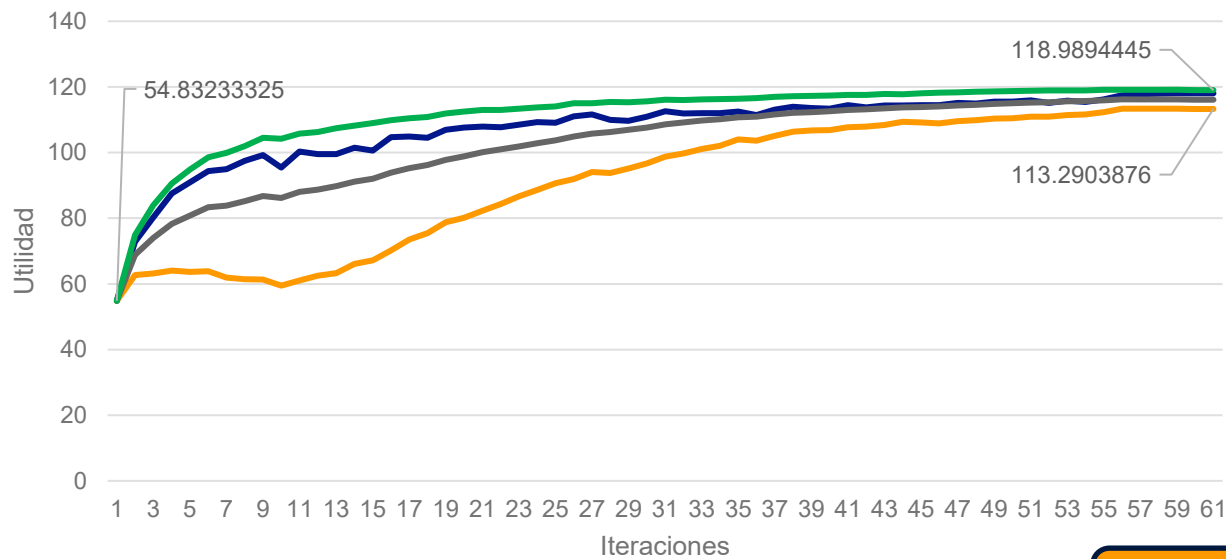
Utilidades - Última Línea 18



Restricción a
una línea

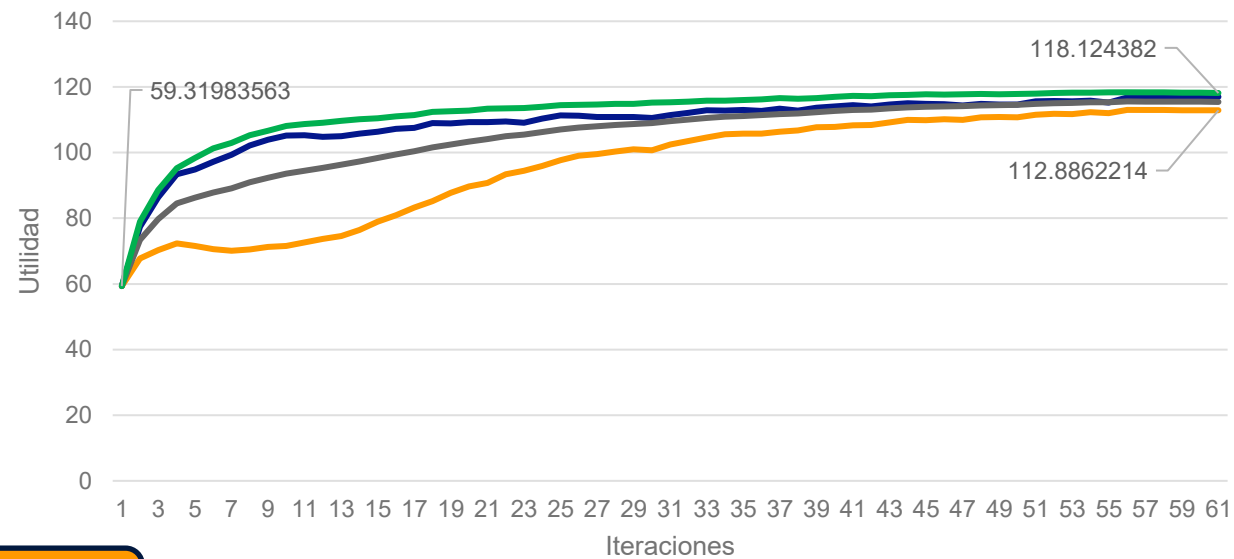
Escenario	Peor Utilidad Iteración 0 (avg_worst)	Mejor Utilidad Iteración 60 (avg_best)	Peor Utilidad Iteración 60 (avg_worst)	Diferencia Iteración 60
Última Línea Tranvía 6	6.86	116.901	107.650	9.251
Última Línea Tranvía 4	-2.567	117.242	103.283	13.959
Última Línea Bus 98	32.329	117.519	110.500	7.019
Última Línea Bus 93	-7.609	116.467	105.330	11.137
Última Línea Bus 71	1.367	116.211	104.347	11.864
Última Línea Bus 40	7.986	116.756	103.559	13.197
Última Línea Bus 18	-6.439	116.285	95.935	20.350

Utilidades - Totalidad de la oferta de Transporte

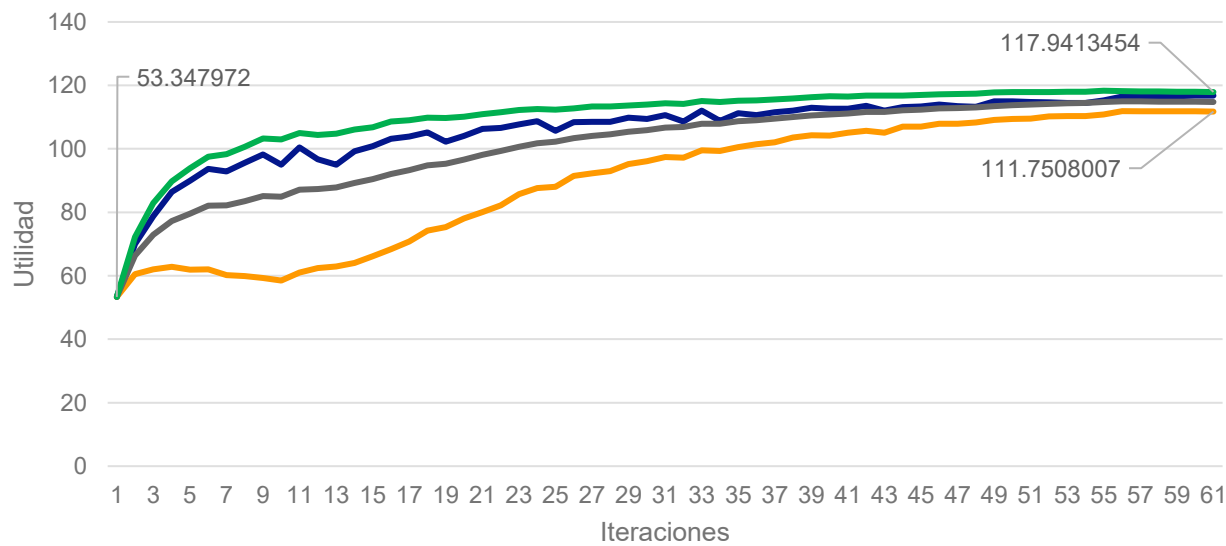


**Restricción
Acumulada**

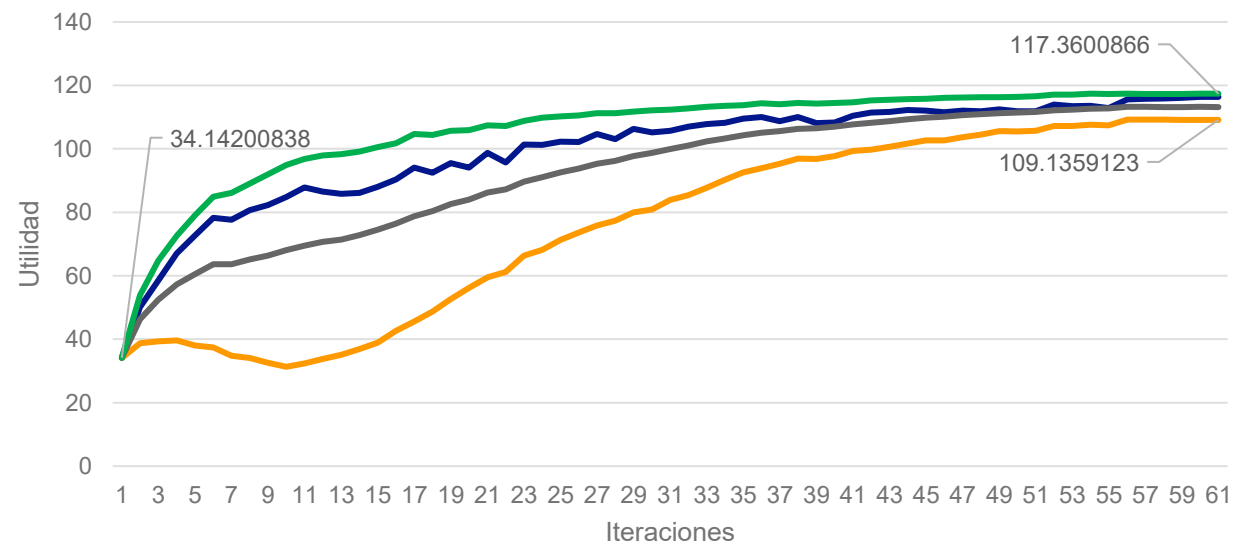
Utilidades - sin Línea Tranvía 6

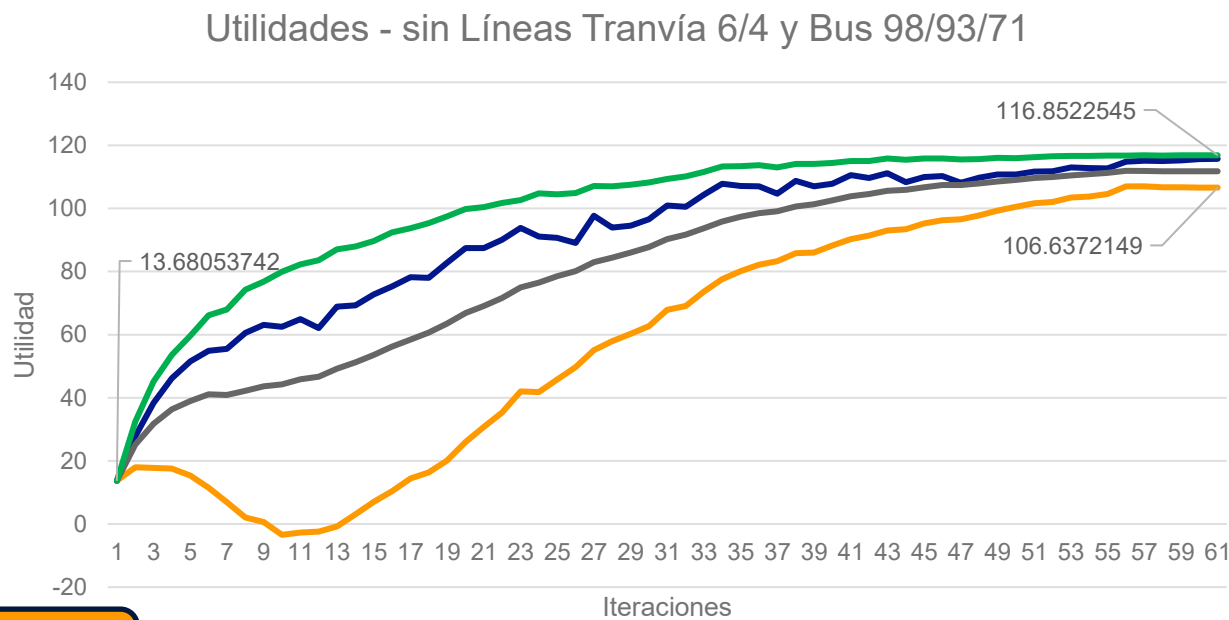
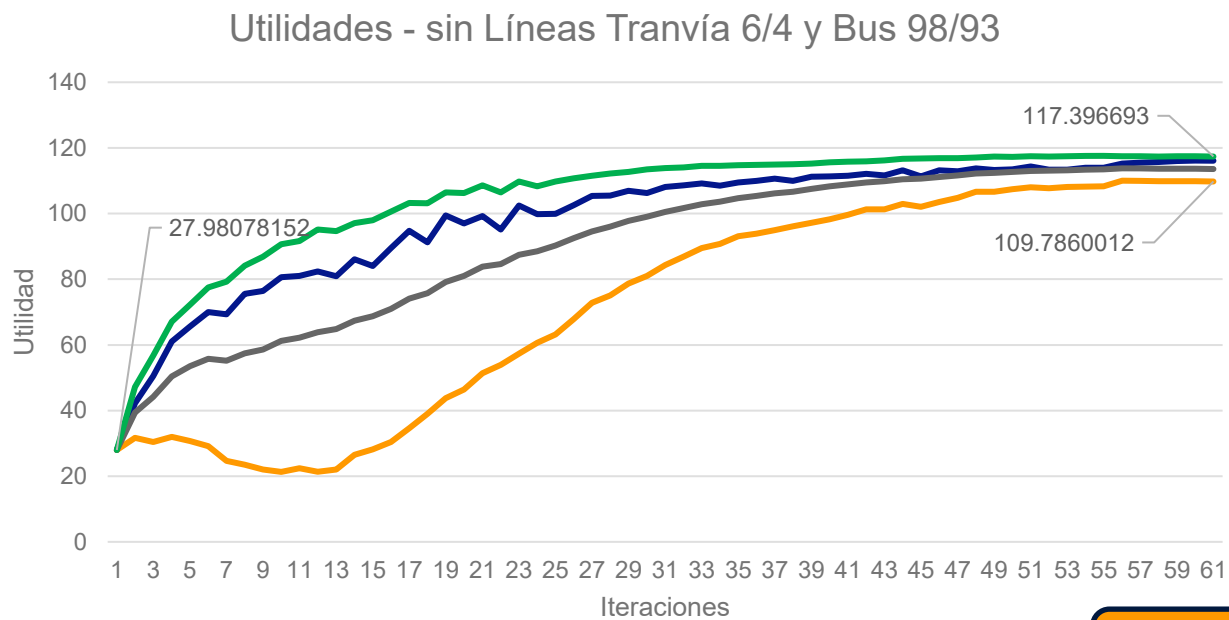


Utilidades - sin Línea Tranvía 6/4



Utilidades - sin Líneas Tranvía 6/4 y Bus 98



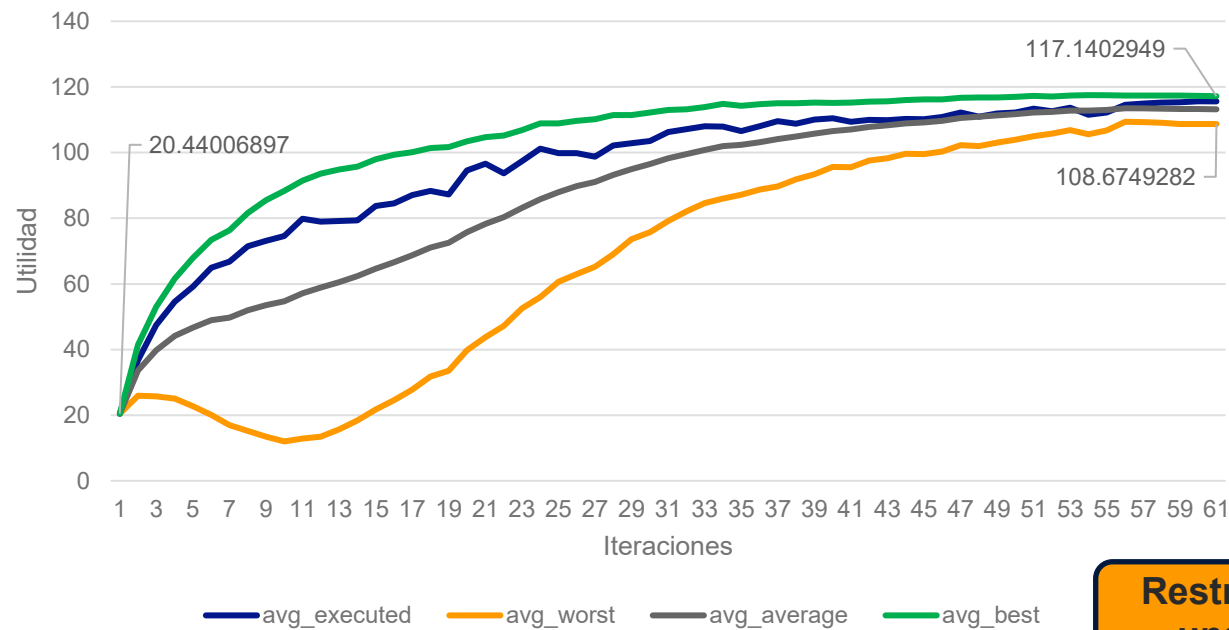


Restricción Acumulada

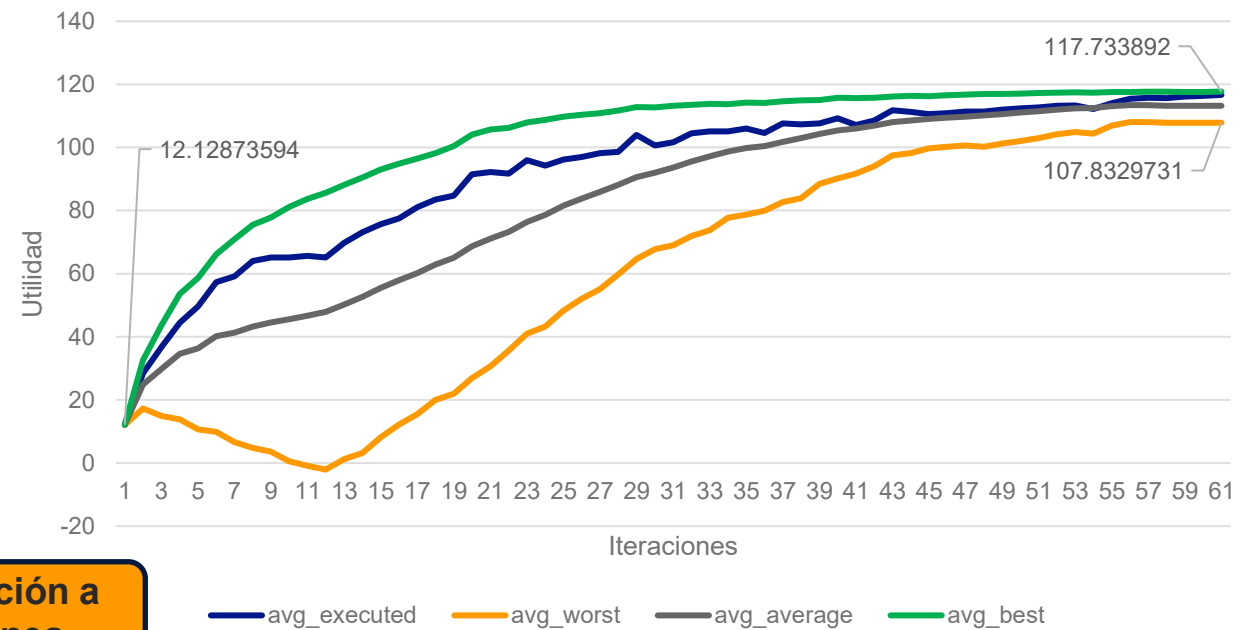
Categoría	Peor Utilidad Iteración 0 (avg_worst)	Mejor Utilidad Iteración 60 (avg_best)	Peor Utilidad Iteración 60 (avg_worst)
Promedio Escenario Base	28.37	117.70	111.01
Promedio Solución	39.65	117.80	110.38
Diferencia Porcentual	39.74%	0.09%	-0.57%

Líneas No simuladas	Líneas Simuladas	Peor Utilidad Iteración 0 (avg_worst)	Mejor Utilidad Iteración 60 (avg_best)	Peor Utilidad Iteración 60 (avg_worst)	Diferencia Iteración 60
Ninguna	Todas	54.832	118.989	113.290	5.699
Tranvía 6	Tranvía 4, Bus 98, 93, 71, 40 y 18	53.907	118.304	111.727	6.577
Tranvía 6 y 4	Bus 98, 93, 71, 40 y 18	53.348	117.941	111.751	6.191
Tranvía 6 y 4, Bus 98	Bus 93, 71, 40 y 18	34.142	117.360	109.136	8.224
Tranvía 6 y 4, Bus 98, 93	Bus 71, 40 y 18	27.981	117.397	109.786	7.611
Tranvía 6 y 4, Bus 98, 93, 71	Bus 40 y 18	13.681	116.852	106.637	10.215

Utilidades - Última Línea Tranvía 6

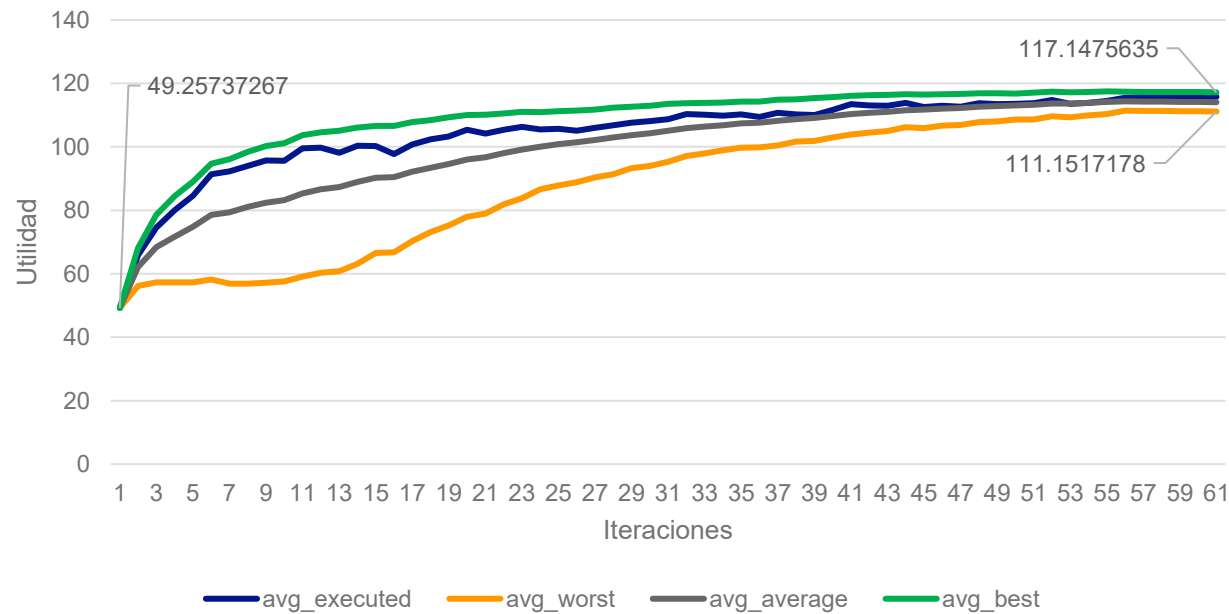


Utilidades - Última Línea Tranvía 4

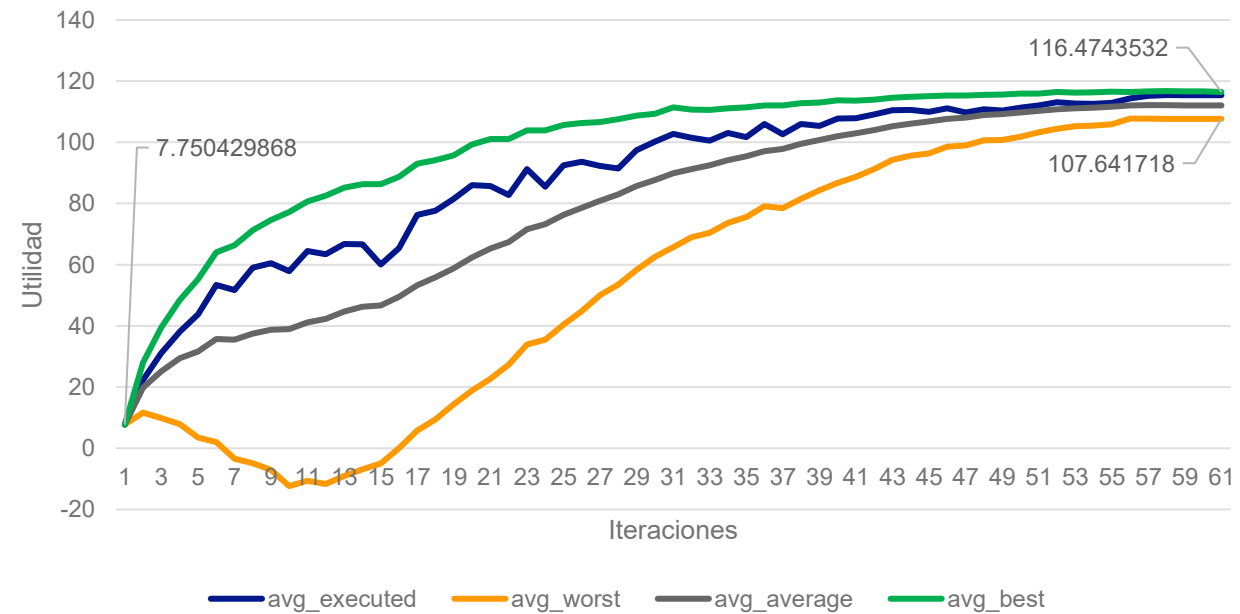


Restricción a una línea

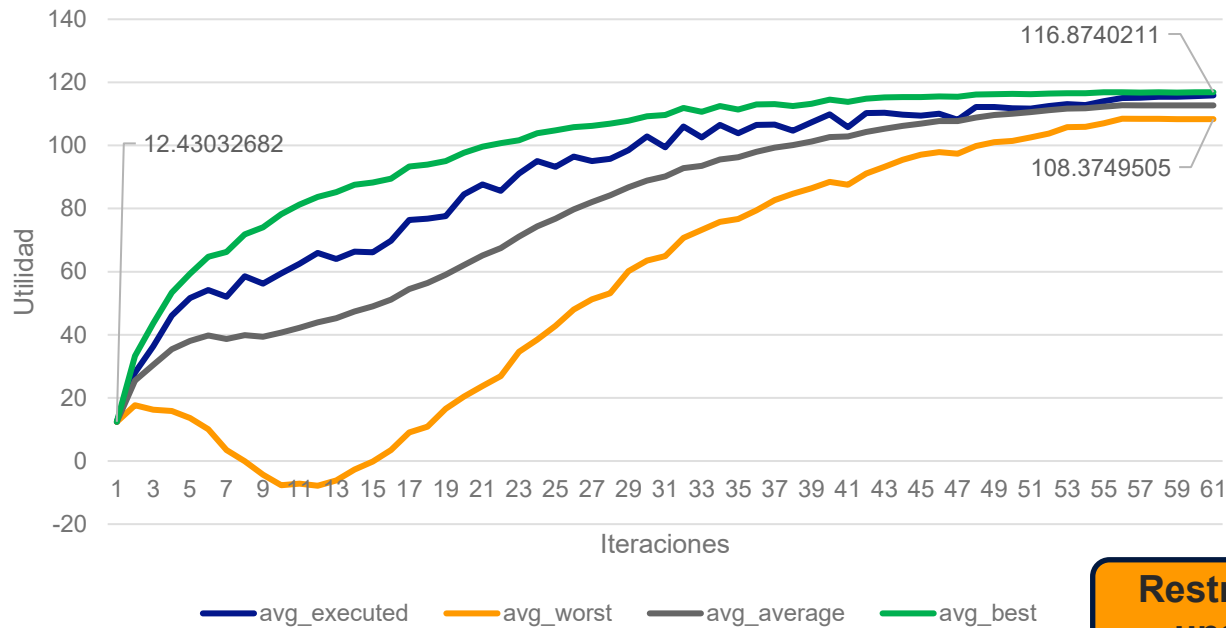
Utilidades - Última Línea Bus 98



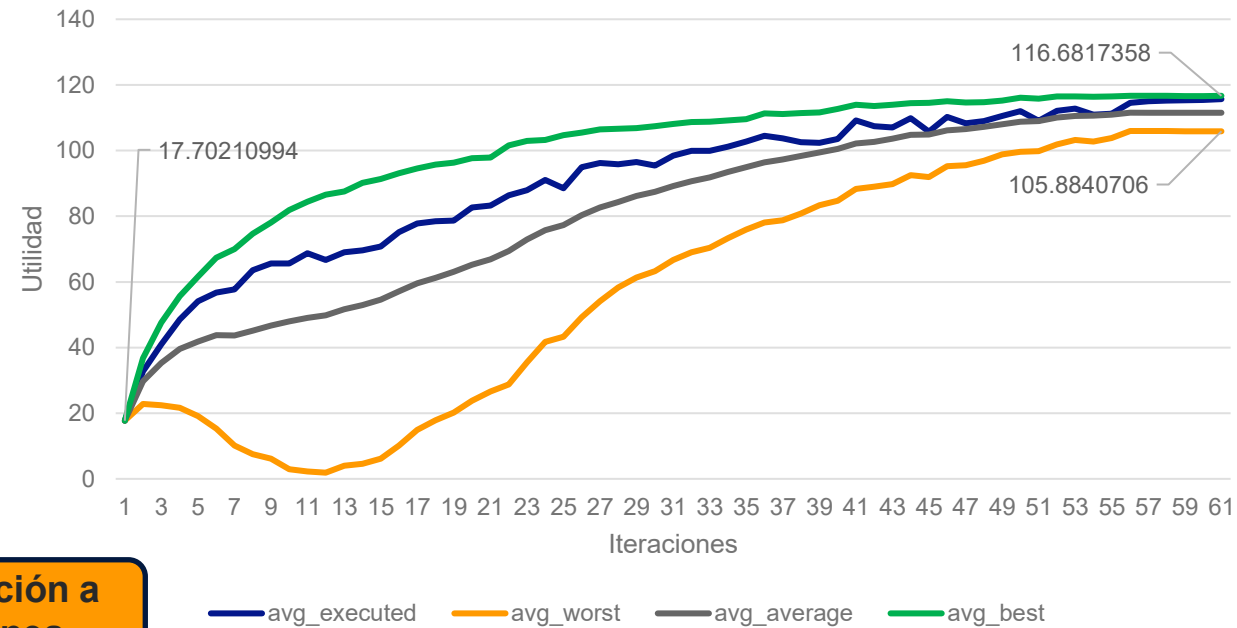
Utilidades - Última Línea Bus 93



Utilidades - Última Línea Bus 71

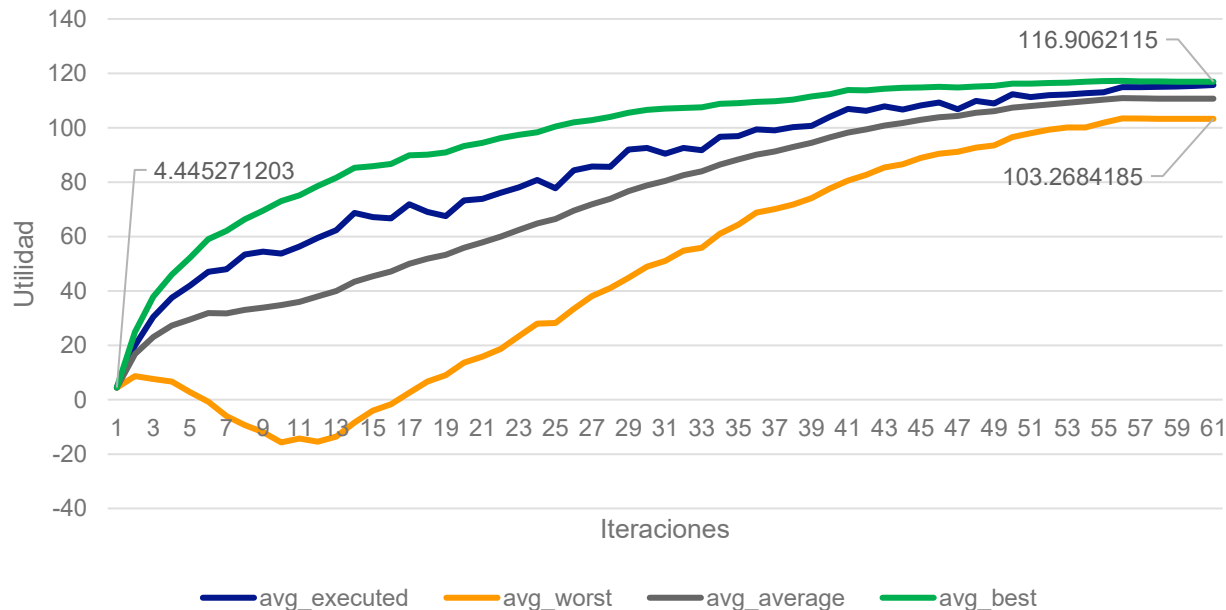


Utilidades - Última Línea Bus 40



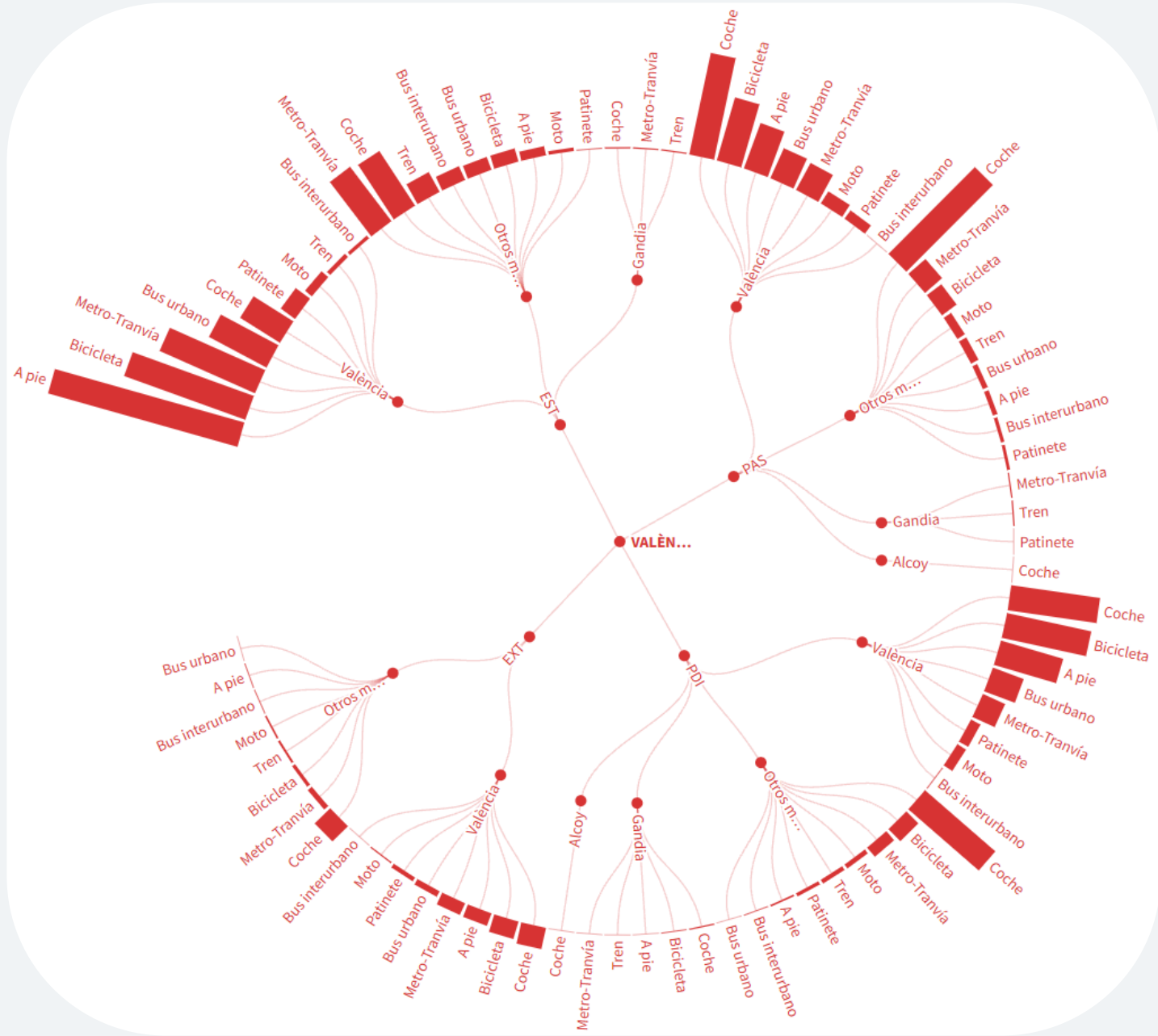
Restricción a una línea

Utilidades - Última Línea 18



Categoría	Peor Utilidad Iteración 0 (avg_worst)	Mejor Utilidad Iteración 60 (avg_best)	Peor Utilidad Iteración 60 (avg_worst)
Promedio Escenario Base	4.561	116.769	104.373
Promedio Solución	17.736	116.994	107.547
Diferencia Porcentual	288.86%	0.19%	3.04%

Escenario	Peor Utilidad Iteración 0 (avg_worst)	Mejor Utilidad Iteración 60 (avg_best)	Peor Utilidad Iteración 60 (avg_worst)	Diferencia Iteración 60
Ultima Línea Tranvía 6	20.440	117.140	108.675	8.465
Ultima Línea Tranvía 4	12.129	117.734	107.833	9.901
Ultima Línea Bus 98	49.257	117.148	111.152	5.996
Ultima Línea Bus 93	7.750	116.474	107.642	8.833
Ultima Línea Bus 71	12.430	116.874	108.375	8.499
Ultima Línea Bus 40	17.702	116.682	105.884	10.798
Ultima Línea Bus 18	4.445	116.906	103.268	13.638





```
<?xml version='1.0' encoding='UTF-8'?>
<!DOCTYPE population SYSTEM "http://www.matsim.org/files/dtd/population_v6.dtd">
```

```
<population>
  <person id="17">
    <attributes>
      <attribute name="carAvail" class="java.lang.String">never</attribute>
    </attributes>
    <plan selected="yes">
      <activity type="h" x="724924.7821" y="4371420.408" end_time="7:03:30"/>
      <leg mode="bicycle"/>
      <activity type="s" x="728351.9221" y="4373419.186" end_time="20:00:00"/>
      <leg mode="bicycle"/>
      <activity type="h" x="724924.7821" y="4371420.408"/>
    </plan>
  </person>
  <person id="20">
    <attributes>
      <attribute name="carAvail" class="java.lang.String">always</attribute>
    </attributes>
    <plan selected="yes">
      <activity type="h" x="724265.6142" y="4371129.895" end_time="8:54:33"/>
      <leg mode="car"/>
      <activity type="w" x="728439.8025" y="4373393.521" end_time="18:00:00"/>
      <leg mode="car"/>
      <activity type="h" x="724265.6142" y="4371129.895"/>
    </plan>
  </person>
  <person id="22">
    <attributes>
      <attribute name="carAvail" class="java.lang.String">always</attribute>
    </attributes>
    <plan selected="yes">
      <activity type="h" x="725605.7121" y="4375248.422" end_time="7:42:45"/>
      <leg mode="car"/>
      <activity type="w" x="728773.9466" y="4373277.133" end_time="19:00:00"/>
      <leg mode="car"/>
      <activity type="h" x="725605.7121" y="4375248.422"/>
    </plan>
  </person>
```

```
<?xml version="1.0" encoding="UTF-8"?>
<vehicleDefinitions xmlns="http://www.matsim.org/files/dtd" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.matsim.org/files/dtd http://www.matsim.org/files/dtd/vehicleDefinitions_v2.0.xsd">
  <vehicleType id="car">
    <attributes>
      <attribute name="accessTimeInSecondsPerPerson" class="java.lang.Double">1.0</attribute>
      <attribute name="doorOperationMode" class="org.matsim.vehicles.VehicleType$DoorOperationMode">serial</attribute>
      <attribute name="egressTimeInSecondsPerPerson" class="java.lang.Double">1.0</attribute>
    </attributes>
    <length meter="7.5"/>
    <width meter="2.0"/>
    <maximumVelocity meterPerSecond="16.67"/>
    <passengerCarEquivalents pce="1.0"/>
  </vehicleType>

  <vehicleType id="bicycle">
    <attributes>
      <attribute name="accessTimeInSecondsPerPerson" class="java.lang.Double">1.0</attribute>
      <attribute name="egressTimeInSecondsPerPerson" class="java.lang.Double">1.0</attribute>
    </attributes>
    <length meter="2"/>
    <width meter="1.0"/>
    <maximumVelocity meterPerSecond="4.17"/>
    <passengerCarEquivalents pce="0.25"/>
  </vehicleType>

  <networkMode networkMode="bicycle"/>
</vehicleDefinitions>
```



```
<transitSchedule>
  <attributes>
    <attribute name="endDate" class="java.lang.String">2022-11-30</attribute>
    <attribute name="startDate" class="java.lang.String">2022-11-30</attribute>
  </attributes>

  <transitStops>
    <stopFacility id="1" x="729197.34" y="4385492.26" linkRefId="pt_205" name="Rafelbunyol" stopAreaId="1" isBlocking="false"/>
    <stopFacility id="1.1" x="729197.34" y="4385492.26" linkRefId="pt_207" name="Rafelbunyol" stopAreaId="1"
isBlocking="false"/>
    <stopFacility id="10" x="727425.11" y="4375122.92" linkRefId="pt_196" name="Alboraia Palmaret" stopAreaId="10"
isBlocking="false"/>
    <stopFacility id="10.1" x="727425.11" y="4375122.92" linkRefId="pt_216" name="Alboraia Palmaret" stopAreaId="10"
isBlocking="false"/>
    <stopFacility id="100" x="723557.09" y="4375178.92" linkRefId="pt_239" name="Palau de Congressos" stopAreaId="100"
isBlocking="false"/>
    <stopFacility id="100.1" x="723557.09" y="4375178.92" linkRefId="pt_266" name="Palau de Congressos" stopAreaId="100"
isBlocking="false"/>

    <transitLine id="1---V1-1-26" name="1">
      <attributes>
        <attribute name="gtfs_agency_id" class="java.lang.String">1</attribute>
        <attribute name="gtfs_route_short_name" class="java.lang.String">1</attribute>
        <attribute name="gtfs_route_type" class="java.lang.String">1</attribute>
      </attributes>

      <transitRoute id="1---V1-1-26_0">
        <attributes>
          <attribute name="simple_route_type" class="java.lang.String">subway</attribute>
        </attributes>

        <transportMode>subway</transportMode>
        <routeProfile>
          <stop refId="55.2" arrivalOffset="00:00:00" departureOffset="00:00:00" awaitDeparture="true"/>
          <stop refId="54" arrivalOffset="00:01:45" departureOffset="00:01:45" awaitDeparture="true"/>
        </routeProfile>
        <route>
          <link refId="pt_90"/>
          <link refId="pt_11"/>
          <link refId="pt_12"/>
          <link refId="pt_13"/>
          <link refId="pt_14"/>
          <link refId="pt_15"/>
        </route>
      </transitRoute>
    </transitLine>
  </transitStops>
</transitSchedule>
```

```
<?xml version='1.0' encoding='UTF-8'?>
```

```
<vehicleDefinitions xmlns="http://www.matsim.org/files/dtd" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.matsim.org/files/dtd http://www.matsim.org/files/dtd/vehicleDefinitions_v2.0.xsd">
```

```
    <vehicleType id="Light_train"> <!-- Vossloh, metro de 4 coches -->
    <attributes>
        <attribute name="accessTimeInSecondsPerPerson" class="java.lang.Double">0.0555555555555555</attribute>
        <attribute name="doorOperationMode" class="org.matsim.vehicles.VehicleType$DoorOperationMode">serial</attribute>
        <attribute name="egressTimeInSecondsPerPerson" class="java.lang.Double">0.0555555555555555</attribute>
    </attributes>
        <capacity seats="2" standingRoomInPersons="9">

        </capacity>
        <length meter="60.49" />
        <width meter="2.55" />
    <costInformation>

</costInformation>
        <passengerCarEquivalents pce="8.0" />
        <networkMode networkMode="pt" />
        <flowEfficiencyFactor factor="1.0" />
    </vehicleType>

    <vehicleType id="tram1"> <!-- Alstom -->
    <attributes>
        <attribute name="accessTimeInSecondsPerPerson" class="java.lang.Double">0.2</attribute>
        <attribute name="doorOperationMode" class="org.matsim.vehicles.VehicleType$DoorOperationMode">serial</attribute>
        <attribute name="egressTimeInSecondsPerPerson" class="java.lang.Double">0.2</attribute>
    </attributes>
        <capacity seats="1" standingRoomInPersons="3">

        </capacity>
        <length meter="23.78" />
        <width meter="2.4" />
    <costInformation>

</costInformation>
        <passengerCarEquivalents pce="3.0" />
        <networkMode networkMode="pt" />
        <flowEfficiencyFactor factor="1.0" />
    </vehicleType>
```

```
<?xml version="1.0" ?>
<!DOCTYPE config SYSTEM "http://www.matsim.org/files/dtd/config_v2.dtd">
<config>

  <module name="controller">
    <param name="outputDirectory" value="./Result_21_s2_2%_TEST" />
    <param name="firstIteration" value="0" />
    <param name="lastIteration" value="60" />
  </module>

  <module name="global">
    <param name="randomSeed" value="4711" />
    <param name="coordinateSystem" value="epsg:25830" />
  </module>

  <module name="network">
    <param name="inputNetworkFile" value="Network_Valencia_Multi_v2.xml" />
  </module>

  <module name="plans">
    <param name="inputPlansFile" value="plans_1971_v6mt_Solu2.xml" />
  </module>

  <module name="vehicles" >
    <param name="vehiclesFile" value="vehiculos.xml" />
  </module>

  <module name="transit"> <!-- TRANSPORTE PUBLICO, DEBE ESTAR EN "TRUE" SI SE QUIERE SIMULAR -->
    <param name="useTransit" value="true"/>
    <param name="transitScheduleFile" value="C:\Users\Rodrigo Medina\Documents\TFM\04_Example
Project\scenarios\equil\Escenario_Valencia\TransitS_v1\transitSchedule_Multi_W_TR6_4_B98_93_71_40.xml"/>
    <param name="vehiclesFile" value="transitVehicles_FGV+EMT_v3.xml"/>
    <param name="transitModes" value="pt"/>
  </module>

  <module name="transitRouter"> <!-- ESTABLECE EL AREA DE BUSQUEDA PARA TOMAR EL TRANSPORTE PUBLICO -->
    <param name="searchRadius" value="250" /> <!-- RADIO INICIAL DE BUSQUEDA DESDE EL PUNTO DE SALIDA DE LA ACTIVIDAD -->
    <param name="extensionRadius" value="500" /> <!-- EXTENSION DEL RADIO POR SI NO ENCUENTRA UNA PARADA / ESTACION -->
    <param name="maxBeelineWalkConnectionDistance" value="200.0" /> <!-- DISTANCIA MAXIMA DE CAMINATA PARA UN POSIBLE TRANSBORDO -->
    <param name="additionalTransferTime" value="180.0" /> <!-- TIEMPO DE TRANSFERENCIA ENTRE TRANSBORDOS. ESTA EN SEGUNDOS -->
    <param name="directWalkFactor" value="1.0" /> <!-- CUANDO ES 1: SI CAMINA POR 5 MIN ENTONCES SERAN 5 MIN -->
  </module>
```